

Mathematical Methods for Finance

Module 1 – Metric, Normed and Hilbert Spaces

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Academic Year 2026/2027

Theory, visual intuition, guided examples, counterexamples and exam-style exercises

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Module 1

Orientation and notation

How to read these slides

Intuition: Four-step routine

For each new concept, use the same order:

- ① picture first,
- ② definition second,
- ③ worked example third,
- ④ counterexample or exercise last.

Exercise: Active reading

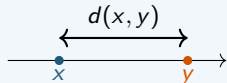
When a slide contains a theorem, underline the hypotheses. Then ask: what breaks if one hypothesis is removed?

Main objective

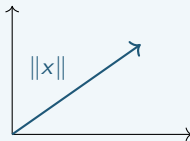
The goal is not to memorize many definitions, but to recognize which definition is useful in a concrete problem.

The three languages of Module 1

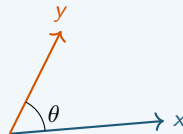
Distance



Length



Angle



Intuition: Translation

Metric spaces speak about distance. Normed spaces speak about vector length. Hilbert spaces add angles and orthogonality.

A practical dictionary

Mathematical word	Practical reading
Metric	How far are two objects?
Convergence	Are approximations approaching a target?
Cauchy	Are approximations stabilizing even before knowing the target?
Completeness	Does the target still belong to the chosen space?
Compactness	Can every infinite search be reduced to a convergent subsequence?
Projection	What is the best approximation inside a simpler set?

Exercise: One-minute check

For each word above, write one sentence beginning with: "This matters because..."

Typical mistake: using a theorem too early

Counterexample: The danger

A theorem is not a formula. Before using it, every hypothesis must be checked.

Unsafe reasoning

- ① I see a sequence.
- ② I expect a limit.
- ③ I conclude the limit exists.

Safe reasoning

- ① Is the sequence Cauchy?
- ② Is the space complete?
- ③ Is the relevant set closed?
- ④ Then the limit belongs to the set.

Exercise: Mini-exam

Give an example where the unsafe reasoning fails.

Running example: approximating a table of numbers

Intuition: Objects as vectors

A finite table with n numerical entries can be viewed as a vector in $\mathbb{R}^n$.

Worked example: Data vector

$$x = (2.1, 1.9, 2.3, 2.0) \in \mathbb{R}^4.$$

A simpler approximation could be the constant vector

$$c(1, 1, 1, 1).$$

Questions

- Which value of c is best?
- What does “best” mean?
- Which distance should be used?

This example will reappear when we discuss projection.

What Module 1 is about

A single idea

Many mathematical problems become clearer once objects are placed in a space with a notion of distance, length or angle.

- Metric spaces: distance, convergence, completeness and compactness.
- Normed spaces: vectors, functions, sequences and linear maps.
- Banach spaces: complete normed spaces where limits of approximations exist.
- Hilbert spaces: geometry in infinite dimension, orthogonality and projection.
- Practical theme: solving equations, approximating objects and controlling errors.

What students should be able to do

- ① Verify whether a proposed formula is a metric, a norm or an inner product.
- ② Determine convergence, Cauchy property, completeness and compactness in examples.
- ③ Use Holder, Minkowski and Cauchy-Schwarz inequalities correctly.
- ④ Compute projections in Euclidean spaces and recognize projection principles in Hilbert spaces.
- ⑤ Prove short statements rigorously and construct counterexamples.
- ⑥ Translate an applied approximation problem into a problem about distance or projection.

Notation used throughout

Sets and spaces

- X, Y : sets or spaces.
- $x_n \rightarrow x$: convergence.
- $B(x, r)$: open ball.
- $\overline{A}$: closure of A .
- A° : interior of A .

Linear notation

- $\|x\|$: norm of x .
- $\langle x, y \rangle$: inner product.
- $x \perp y$: orthogonality.
- $T : X \rightarrow Y$: linear map.
- $\|T\|$: operator norm.

A recurring pattern

Three questions

When a sequence of approximations (x_n) is constructed, always ask:

- ① Is (x_n) Cauchy? This means the approximations are stabilizing.
- ② Does the ambient space contain its Cauchy limits? This is completeness.
- ③ Is the limit the desired object? This requires continuity or closedness.

Example: Practical meaning

If a numerical method generates better and better approximations, completeness is the mathematical property that prevents the limit from falling outside the chosen space.

How exam questions will look

Exercise: Typical structure

A good question often combines four tasks:

- ① State the relevant definition.
- ② Verify hypotheses carefully.
- ③ Apply a theorem or construct a counterexample.
- ④ Interpret the conclusion in one clear sentence.

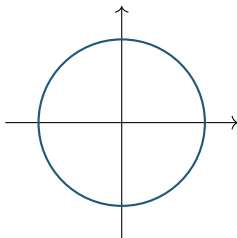
Warning

Most mistakes in this module come from using the right theorem in a space where one hypothesis is missing: completeness, closedness, boundedness or compactness.

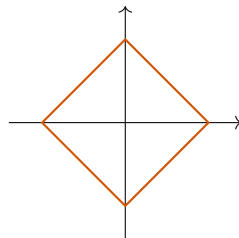
Module 1

Metric spaces

Visual intuition: metric balls in the plane



d_2 ball: round

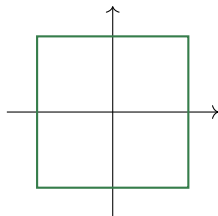


d_1 ball: diamond

Intuition: Same center, different geometry

Changing the distance changes the shape of neighborhoods, hence the way we visualize convergence.

Visual intuition: the sup metric



d_∞ ball: square

Meaning in $\mathbb{R}^2$

$$d_\infty(x, y) = \max\{|x_1 - y_1|, |x_2 - y_2|\}.$$

A point is inside the ball if every coordinate error is less than the radius.

Exercise: Quick task

Draw the ball $B_\infty((1, 1), 2)$ in $\mathbb{R}^2$.

Worked example: comparing three metrics

Let $x = (1, 2)$ and $y = (4, 6)$.

Worked example: d_1

$$d_1(x, y) = |1-4| + |2-6| = 7.$$

Worked example: d_2

$$d_2(x, y) = \sqrt{3^2 + 4^2} = 5.$$

Worked example: d_∞

$$d_\infty(x, y) = \max\{3, 4\} = 4.$$

Observation

All three are valid distances, but they measure errors differently.

Exercise: verify a metric step by step

Exercise: Candidate distance

On $\mathbb{R}$, define

$$d(x, y) = \frac{|x - y|}{1 + |x - y|}.$$

Show that d is a metric.

Suggested structure

- 1 Non-negativity and identity: use $|x - y| \geq 0$.
- 2 Symmetry: use $|x - y| = |y - x|$.
- 3 Triangle inequality: use the function $\varphi(t) = t/(1 + t)$.

Intuition: Why this is useful

This metric compresses large distances: all distances are less than 1.

Solution skeleton: the bounded metric

Proof idea: Triangle inequality

Let $a, b \geq 0$. We need

$$\frac{a+b}{1+a+b} \leq \frac{a}{1+a} + \frac{b}{1+b}.$$

This follows because $\varphi(t) = t/(1+t)$ is increasing and subadditive. Then

$$|x - z| \leq |x - y| + |y - z|$$

implies

$$d(x, z) \leq d(x, y) + d(y, z).$$

Exercise: Complete the proof

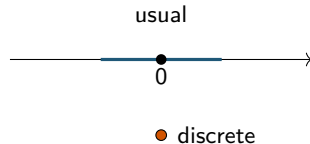
Prove directly that $\varphi(a+b) \leq \varphi(a) + \varphi(b)$.

Open balls: always ask which metric

Worked example: Same set, two metrics

On $\mathbb{R}$, take the point 0 and radius $1/2$.

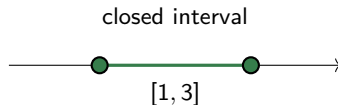
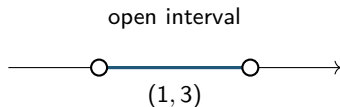
- Usual metric: $B(0, 1/2) = (-1/2, 1/2)$.
- Discrete metric: $B(0, 1/2) = \{0\}$.



Exercise: Check

What is $B(0, 2)$ under the discrete metric?

Open and closed sets: a visual reminder



Reminder

Open and closed are topological properties: they depend on the ambient space.

Relative topology: the ambient space matters

Worked example: A set can change status

Consider $A = (0, 1)$.

- As a subset of $\mathbb{R}$, A is open but not closed.
- As a subset of $X = (0, 1)$, $A = X$ is both open and closed.

Counterexample: Common mistake

Do not say simply “ A is open” without knowing the ambient space.

Exercise: Mini-exam

In $X = [0, 1]$ with the usual metric inherited from $\mathbb{R}$, is $[0, 1/2)$ open in X ?

Solution: relative openness

Proof idea: Answer

In $X = [0, 1]$, the set $[0, 1/2)$ is open because

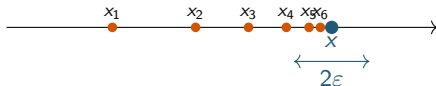
$$[0, 1/2) = X \cap (-1/2, 1/2).$$

It is the intersection of the ambient space X with an open set of $\mathbb{R}$.

Intuition: Geometric reading

Inside $X = [0, 1]$, the point 0 is not a boundary obstruction on the left: there are no points of X below 0.

Convergence: moving target picture



Intuition: Definition in words

For every tolerance ε , after some index N , all terms remain inside the ε -ball around x .

Exercise: Explain aloud

Why does the value of N usually depend on ε ?

Exercise: convergence in different metrics

Let $x_n = 1/n$ on $\mathbb{R}$.

Exercise: Two questions

- 1 Does $x_n \rightarrow 0$ under the usual metric?
- 2 Does $x_n \rightarrow 0$ under the metric $d(x, y) = \min\{1, |x - y|\}$?

Hint

For small distances, $\min\{1, |x - y|\} = |x - y|$.

Worked example: Answer

Yes in both cases. The second metric is bounded, but it agrees with the usual metric near zero.

Cauchy without knowing the limit

Intuition: Convergent sequence

The terms approach a known point x .

$$d(x_n, x) \rightarrow 0.$$

Intuition: Cauchy sequence

The terms approach each other.

$$d(x_n, x_m) \rightarrow 0 \quad (m, n \rightarrow \infty).$$

Exercise: Concept check

Why can a Cauchy sequence fail to converge inside the space?

Practical example: stopping an iterative algorithm

Worked example: Stopping rule

Suppose an algorithm produces $x_1, x_2, \dots$ in a metric space. A common stopping rule is

$$d(x_{n+1}, x_n) < 10^{-6}.$$

This checks that consecutive terms are close, but not necessarily that the whole tail is Cauchy.

Counterexample: Warning

Small consecutive jumps do not always imply convergence. Example on $\mathbb{R}$:

$$x_n = \log n, \quad x_{n+1} - x_n \rightarrow 0,$$

but $x_n \rightarrow +\infty$.

Exam-style exercise: topology of a set

Exercise: Full classification

Let

$$A = (0, 1] \cup \{2\} \subset \mathbb{R}.$$

Find:

- ① the interior A° ;
- ② the closure $\overline{A}$;
- ③ the boundary ∂A ;
- ④ the isolated points;
- ⑤ the limit points.

Answer outline

Draw the set on the real line before writing formulas.

Solution: topology of $(0, 1] \cup \{2\}$

Worked example: Classification

For $A = (0, 1] \cup \{2\}$:

$$A^\circ = (0, 1), \quad \overline{A} = [0, 1] \cup \{2\},$$

$$\partial A = \{0, 1, 2\}.$$

The point 2 is isolated. The limit points are all points of $[0, 1]$.

Intuition: Why 2 lies on the boundary

Every small ball around 2 meets A at 2 and also meets the complement of A .

Metric spaces: definition

Definition: Metric

Let X be a set. A function $d : X \times X \rightarrow [0, +\infty)$ is a metric if, for all $x, y, z \in X$,

$$d(x, y) = 0 \iff x = y,$$

$$d(x, y) = d(y, x),$$

$$d(x, z) \leq d(x, y) + d(y, z).$$

The pair (X, d) is called a metric space.

Interpretation

A metric is an abstract rule for measuring distance. It does not need coordinates, derivatives or algebraic operations.

First examples of metrics

Example: Standard metrics

- ① On $\mathbb{R}$: $d(x, y) = |x - y|$.
- ② On $\mathbb{R}^n$: $d_2(x, y) = \left(\sum_{i=1}^n |x_i - y_i|^2\right)^{1/2}$.
- ③ On any set X : the discrete metric

$$d(x, y) = \begin{cases} 0, & x = y, \\ 1, & x \neq y. \end{cases}$$

- ④ On continuous functions: $d_\infty(f, g) = \sup_{t \in [a, b]} |f(t) - g(t)|$.

Why the three metric axioms matter

Identity

$d(x, y) = 0$ only when $x = y$.

Otherwise different points could be indistinguishable.

Symmetry

$d(x, y) = d(y, x)$.

Distance does not depend on direction.

Triangle

$d(x, z) \leq d(x, y) + d(y, z)$.

Going directly cannot be longer than going through an intermediate point.

Exercise: Quick check

Which axiom fails for $d(x, y) = x - y$ on $\mathbb{R}$?

Counterexamples to metric axioms

Counterexample: Not every distance-like formula is a metric

- ① $d(x, y) = (x - y)^2$ on $\mathbb{R}$ is nonnegative and symmetric, but fails the triangle inequality.
- ② $d(x, y) = |x^2 - y^2|$ on $\mathbb{R}$ fails identity: $d(1, -1) = 0$ but $1 \neq -1$.
- ③ $d(x, y) = |x - y|/(1 + |x - y|)$ is a metric, but this must be proved; boundedness alone is not a problem.

Exercise: Exam-style

Show explicitly that $(x - y)^2$ fails the triangle inequality by choosing three points.

Solution: squared distance is not a metric

Proof idea: A single counterexample

Choose $x = 0$, $y = 1$, $z = 2$. Then

$$d(0, 2) = (0 - 2)^2 = 4.$$

But

$$d(0, 1) + d(1, 2) = 1^2 + 1^2 = 2.$$

The triangle inequality would require $4 \leq 2$, which is false.

Lesson

To disprove a universal property, one counterexample is enough.

Metrics generated by transformations

Theorem: A useful trick

If (X, d) is a metric space and $\varphi : [0, +\infty) \rightarrow [0, +\infty)$ is increasing, $\varphi(0) = 0$ and

$$\varphi(a + b) \leq \varphi(a) + \varphi(b),$$

then

$$\rho(x, y) = \varphi(d(x, y))$$

is a metric.

Example: Bounded metric

With $\varphi(t) = t/(1 + t)$,

$$\rho(x, y) = \frac{d(x, y)}{1 + d(x, y)}$$

is a bounded metric.

Open balls

Definition: Open ball

In a metric space (X, d) , the open ball centered at $x \in X$ with radius $r > 0$ is

$$B(x, r) = \{y \in X : d(x, y) < r\}.$$

Example: On the real line

With $d(x, y) = |x - y|$,

$$B(2, 0.5) = (1.5, 2.5).$$

Example: Discrete metric

If $0 < r \leq 1$, then $B(x, r) = \{x\}$; if $r > 1$, then $B(x, r) = X$.

Open sets

Definition: Open set

A set $A \subset X$ is open if for every $x \in A$ there exists $r > 0$ such that

$$B(x, r) \subset A.$$

Example: Intervals

In $\mathbb{R}$, $(0, 1)$ is open. For $x \in (0, 1)$ choose

$$r = \frac{1}{2} \min\{x, 1 - x\}.$$

Then $B(x, r) \subset (0, 1)$.

Closed sets

Definition: Closed set

A set $F \subset X$ is closed if its complement $X \setminus F$ is open.

Theorem: Sequential characterization

In a metric space, F is closed if and only if every convergent sequence (x_n) with $x_n \in F$ and $x_n \rightarrow x$ has $x \in F$.

Example: Example

$[0, 1]$ is closed in $\mathbb{R}$. If $x_n \in [0, 1]$ and $x_n \rightarrow x$, then $0 \leq x \leq 1$.

Sets can be both open and closed

Counterexample: Common misconception

A set can be both open and closed.

Example: In every metric space

$\emptyset$ and X are both open and closed.

Example: Discrete metric

In the discrete metric, every subset $A \subset X$ is open. Therefore every subset is also closed.

Lesson

“Open” does not mean “not closed”. The two properties are not opposites.

Interior, closure and boundary

Definition: Three associated sets

Let $A \subset X$.

- The interior A° is the largest open set contained in A .
- The closure $\bar{A}$ is the smallest closed set containing A .
- The boundary is $\partial A = \bar{A} \setminus A^\circ$.

Example: In $\mathbb{R}$

For $A = (0, 1] \cup \{2\}$,

$$\begin{aligned} A^\circ &= (0, 1), & \bar{A} &= [0, 1] \cup \{2\}, \\ \partial A &= \{0, 1, 2\}. \end{aligned}$$

Limit points and isolated points

Definition: Limit point

A point $x \in X$ is a limit point of A if every ball $B(x, r)$ contains a point of A different from x .

Example: Set with an isolated point

For $A = \{1/n : n \in \mathbb{N}\} \cup \{2\} \subset \mathbb{R}$:

- 0 is a limit point of A .
- 2 is an isolated point.
- Each $1/n$ is isolated, although the sequence accumulates at 0.

Convergence in a metric space

Definition: Convergence

A sequence (x_n) in (X, d) converges to $x \in X$ if

$$\forall \varepsilon > 0 \exists N \in \mathbb{N} \text{ such that } n \geq N \Rightarrow d(x_n, x) < \varepsilon.$$

We write $x_n \rightarrow x$.

Meaning

Eventually all elements of the sequence are inside every ball centered at the limit.

Uniqueness of limits

Theorem: Limits are unique

In a metric space, a convergent sequence has at most one limit.

Proof idea: Using the triangle inequality

Assume $x_n \rightarrow x$ and $x_n \rightarrow y$. Then

$$d(x, y) \leq d(x, x_n) + d(x_n, y).$$

For every $\varepsilon > 0$, for n large enough both terms are less than $\varepsilon/2$. Hence

$$d(x, y) < \varepsilon.$$

Since this holds for every $\varepsilon > 0$, $d(x, y) = 0$, so $x = y$.

Example: convergence in $\mathbb{R}^2$

Example: A sequence of points

Let

$$x_n = \left(\frac{1}{n}, \frac{(-1)^n}{n} \right) \in \mathbb{R}^2.$$

With the Euclidean metric,

$$d_2(x_n, 0) = \sqrt{\frac{1}{n^2} + \frac{1}{n^2}} = \frac{\sqrt{2}}{n} \rightarrow 0.$$

Therefore $x_n \rightarrow (0, 0)$.

Exercise: Variation

Does $y_n = (1/n, (-1)^n)$ converge in $\mathbb{R}^2$?

Convergence depends on the metric

Counterexample: Same set, different metric

On $\mathbb{R}$, let d be the usual metric and let d_{disc} be the discrete metric. The sequence $x_n = 1/n$ satisfies

$$x_n \rightarrow 0 \quad \text{under } d.$$

But under d_{disc} ,

$$d_{disc}(x_n, 0) = 1 \quad \text{for every } n,$$

so x_n does not converge to 0.

Lesson

A sequence converges relative to a chosen metric, not in isolation.

Cauchy sequences

Definition: Cauchy sequence

A sequence (x_n) in (X, d) is Cauchy if

$$\forall \varepsilon > 0 \exists N \in \mathbb{N} \text{ such that } m, n \geq N \Rightarrow d(x_n, x_m) < \varepsilon.$$

Meaning

The terms eventually become close to each other. The definition does not mention a candidate limit.

Theorem: Basic fact

Every convergent sequence is Cauchy.

Proof: convergence implies Cauchy

Proof idea: Triangle inequality again

Suppose $x_n \rightarrow x$. For $\varepsilon > 0$, choose N such that

$$n \geq N \Rightarrow d(x_n, x) < \varepsilon/2.$$

Then, if $m, n \geq N$,

$$d(x_n, x_m) \leq d(x_n, x) + d(x, x_m) < \varepsilon.$$

Therefore (x_n) is Cauchy.

Important

The converse requires completeness of the space.

Completeness

Definition: Complete metric space

A metric space (X, d) is complete if every Cauchy sequence in X converges to a point of X .

Example: Complete spaces

- $\mathbb{R}$ with the usual metric.
- $\mathbb{R}^n$ with the Euclidean metric.
- A closed subset of a complete metric space.

Counterexample: Incomplete spaces

$\mathbb{Q}$ is not complete with the usual metric.

Why $\mathbb{Q}$ is not complete

Counterexample: A Cauchy sequence without rational limit

Let (q_n) be rational decimal approximations of $\sqrt{2}$:

$$1, \frac{14}{10}, \frac{141}{100}, \frac{1414}{1000}, \dots$$

This sequence is Cauchy in $\mathbb{Q}$ because it converges in $\mathbb{R}$. But its real limit is $\sqrt{2} \notin \mathbb{Q}$.

Lesson

A sequence may converge in a larger space while failing to converge inside the space under study.

Closed subspaces of complete spaces

Theorem: Closed subset theorem

Let (X, d) be complete and let $F \subset X$. Then F is complete with the inherited metric if and only if F is closed in X .

Proof idea: One direction

If F is closed and (x_n) is Cauchy in F , then it is Cauchy in X . Since X is complete, $x_n \rightarrow x \in X$. Since F is closed and $x_n \in F$, $x \in F$.

Exercise: complete or not?

Exercise: Classify the spaces

With the usual metric inherited from $\mathbb{R}$, decide whether each set is complete.

- ① $[0, 1]$.
- ② $(0, 1)$.
- ③ $\{0\} \cup \{1/n : n \in \mathbb{N}\}$.
- ④ $\mathbb{R} \setminus \{0\}$.

Hint

In $\mathbb{R}$, completeness of a subset is equivalent to closedness.

Solution: complete or not?

Answers

- ① $[0, 1]$ is complete: it is closed in $\mathbb{R}$.
- ② $(0, 1)$ is not complete: $x_n = 1/n$ is Cauchy but has limit $0 \notin (0, 1)$.
- ③ $\{0\} \cup \{1/n : n \in \mathbb{N}\}$ is complete: it is closed in $\mathbb{R}$.
- ④ $\mathbb{R} \setminus \{0\}$ is not complete: $x_n = 1/n$ is Cauchy and converges to 0.

Continuity in metric spaces

Definition: Continuity at a point

Let (X, d_X) and (Y, d_Y) be metric spaces. A function $f : X \rightarrow Y$ is continuous at $x \in X$ if

$$\forall \varepsilon > 0 \exists \delta > 0 \text{ such that } d_X(x, z) < \delta \Rightarrow d_Y(f(x), f(z)) < \varepsilon.$$

Theorem: Sequential characterization

f is continuous at x if and only if

$$x_n \rightarrow x \quad \Rightarrow \quad f(x_n) \rightarrow f(x).$$

Uniform continuity

Definition: Uniform continuity

A function $f : X \rightarrow Y$ is uniformly continuous if

$$\forall \varepsilon > 0 \exists \delta > 0 \text{ such that } d_X(x, z) < \delta \Rightarrow d_Y(f(x), f(z)) < \varepsilon$$

for all $x, z \in X$.

Difference

Continuity allows δ to depend on the point. Uniform continuity uses one δ for the whole domain.

Counterexample: Standard example

$f(x) = x^2$ is continuous on $\mathbb{R}$ but not uniformly continuous on $\mathbb{R}$.

Lipschitz functions

Definition: Lipschitz continuity

A function $f : X \rightarrow Y$ is Lipschitz if there exists $L \geq 0$ such that

$$d_Y(f(x), f(z)) \leq L d_X(x, z)$$

for every $x, z \in X$.

Theorem: Lipschitz implies uniformly continuous

If f is Lipschitz, choose $\delta = \varepsilon/L$ when $L > 0$.

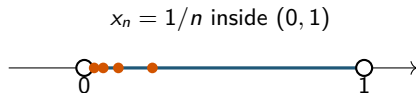
Example: Example

$f(x) = 3x + 2$ is Lipschitz on $\mathbb{R}$ with $L = 3$.

Module 1

Completeness, contraction and compactness

Completeness: the missing-point picture



Counterexample: Incomplete space

The sequence $x_n = 1/n$ is Cauchy in $(0, 1)$ but its limit is 0, which is outside the space.

Moral

Completeness is about not losing limits at the boundary.

Completeness checklist

How to decide whether a subset of $\mathbb{R}^n$ is complete

With the usual metric:

- ① Is the ambient space $\mathbb{R}^n$ complete? Yes.
- ② Is the subset closed in $\mathbb{R}^n$?
- ③ If yes, the subset is complete. If no, usually build a Cauchy sequence converging to a missing boundary point.

Exercise: Apply the checklist

Classify the following subsets of $\mathbb{R}^2$:

$$\{(x, y) : x^2 + y^2 \leq 1\}, \quad \{(x, y) : x^2 + y^2 < 1\}.$$

Solution: closed disk and open disk

Worked example: Closed disk

$$D = \{(x, y) : x^2 + y^2 \leq 1\}$$

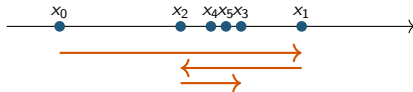
is closed in $\mathbb{R}^2$, hence complete.

Counterexample: Open disk

$$U = \{(x, y) : x^2 + y^2 < 1\}$$

is not complete. The sequence $x_n = (1 - 1/n, 0)$ is Cauchy but converges to $(1, 0) \notin U$.

Contraction: visual intuition



Intuition: Iteration

If T is a contraction, the iteration

$$x_{n+1} = T(x_n)$$

forces points closer and closer to the unique fixed point.

Exercise: Question

Which hypothesis guarantees that the limiting point belongs to the space?

Worked example: a contraction on $\mathbb{R}$

Worked example: Affine map

Let $T(x) = \frac{1}{2}x + 1$ on $\mathbb{R}$.

$$|T(x) - T(y)| = \frac{1}{2}|x - y|.$$

So T is a contraction with constant $q = 1/2$.

Fixed point

Solve $x = T(x)$:

$$x = \frac{1}{2}x + 1 \quad \Rightarrow \quad x = 2.$$

Starting from any x_0 , the iteration $x_{n+1} = T(x_n)$ converges to 2.

Exercise: contraction or not?

Exercise: Classify

On $\mathbb{R}$ with the usual metric, decide whether each map is a contraction.

- ① $T(x) = 0.8x + 3$.
- ② $T(x) = x/2 + \sin x$.
- ③ $T(x) = x^2$ on $[0, 1/4]$.
- ④ $T(x) = \cos x$ on $\mathbb{R}$.

Hint

A differentiable map on an interval is a contraction if $\sup |T'(x)| < 1$.

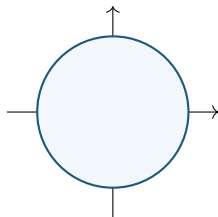
Solution: contraction or not?

- ① $T(x) = 0.8x + 3$: contraction with constant 0.8.
- ② $T(x) = x/2 + \sin x$: not a contraction on $\mathbb{R}$ because $T'(x) = 1/2 + \cos x$ can be $3/2$.
- ③ $T(x) = x^2$ on $[0, 1/4]$: $|T'(x)| = 2x \leq 1/2$, so contraction.
- ④ $T(x) = \cos x$: $|T'(x)| = |\sin x| \leq 1$ but not < 1 uniformly. This argument does not prove contraction.

Counterexample: Important

Having $|T'| \leq 1$ is not enough for Banach's fixed point theorem.

Compactness: finite-dimensional intuition



closed and bounded

Intuition: In $\mathbb{R}^n$

A set is compact if it is closed and bounded.

Compactness prevents:

- escaping to infinity;
- approaching a missing boundary point.

Counterexample: Warning

Closed and bounded is not enough in every metric space.

Exercise: compact or not in $\mathbb{R}$?

Exercise: Classify

With the usual metric, decide whether each set is compact.

- ① $[0, 1]$.
- ② $(0, 1)$.
- ③ $[0, +\infty)$.
- ④ $\{0\} \cup \{1/n : n \in \mathbb{N}\}$.

Use Heine-Borel

In $\mathbb{R}$, compact means closed and bounded.

Solution: compactness in $\mathbb{R}$

- ① $[0, 1]$ is compact: closed and bounded.
- ② $(0, 1)$ is not compact: not closed.
- ③ $[0, +\infty)$ is not compact: not bounded.
- ④ $\{0\} \cup \{1/n : n \in \mathbb{N}\}$ is compact: closed and bounded.

Intuition: The last case

The only possible accumulation point is 0, and it is included in the set.

Sequential compactness: what it says

Theorem: Metric spaces

In metric spaces, compactness is equivalent to sequential compactness: every sequence has a convergent subsequence whose limit belongs to the set.

Compact set

Any infinite list contains a convergent sub-list.

Counterexample: Non-compact set

There exists at least one sequence with no convergent subsequence in the set.

Counterexample: closed and bounded can fail

Counterexample: In infinite dimension

In the space ℓ^2 , the set

$$\{e_1, e_2, e_3, \dots\}$$

is bounded, where e_n has 1 in the n -th coordinate and 0 elsewhere. But

$$\|e_n - e_m\|_2 = \sqrt{2} \quad (n \neq m),$$

so no subsequence can be Cauchy.

Lesson

Infinite-dimensional compactness is much stronger than closed and bounded.

Contractions

Definition: Contraction

Let (X, d) be a metric space. A map $T : X \rightarrow X$ is a contraction if there exists $q \in [0, 1)$ such that

$$d(Tx, Ty) \leq qd(x, y)$$

for all $x, y \in X$.

Why this matters

Repeated iteration

$$x_{n+1} = T(x_n)$$

should bring points closer and closer together.

Banach fixed point theorem

Theorem: Banach contraction theorem

Let (X, d) be a complete metric space and let $T : X \rightarrow X$ be a contraction. Then:

- 1 There exists a unique $x^* \in X$ such that $T(x^*) = x^*$.
- 2 For every starting point $x_0 \in X$, the iteration $x_{n+1} = T(x_n)$ converges to x^* .

Proof idea: Banach fixed point

Proof idea: Four steps

- 1 Show that the iterates satisfy

$$d(x_{n+1}, x_n) \leq q^n d(x_1, x_0).$$

- 2 Sum a geometric series to prove (x_n) is Cauchy.
- 3 Completeness gives $x_n \rightarrow x^* \in X$.
- 4 Continuity of T gives $T(x^*) = x^*$.
- 5 If $Ty = y$, then

$$d(x^*, y) = d(Tx^*, Ty) \leq qd(x^*, y),$$

so $x^* = y$.

Example: solving an equation by contraction

Example: Fixed point form

Solve

$$x = \cos x.$$

Let $T(x) = \cos x$ on $[0, 1]$.

- $T([0, 1]) \subset [0, 1]$.
- $|T'(x)| = |\sin x| \leq \sin 1 < 1$ on $[0, 1]$.

Therefore T is a contraction on the complete metric space $[0, 1]$.

Conclusion

The iteration $x_{n+1} = \cos x_n$ converges to the unique solution in $[0, 1]$.

Counterexample: why completeness is needed

Counterexample: Contraction without fixed point

Let $X = (0, 1)$ and define

$$T(x) = \frac{x}{2}.$$

Then

$$|T(x) - T(y)| = \frac{1}{2}|x - y|,$$

so T is a contraction. But the only possible fixed point is $x = 0$, which is not in X .

Lesson

The theorem fails because $(0, 1)$ is not complete.

Boundedness

Definition: Bounded set

A set $A \subset X$ is bounded if there exist $x_0 \in X$ and $R > 0$ such that

$$A \subset B(x_0, R).$$

Example: In $\mathbb{R}$

$(0, 1)$ is bounded. The set $\mathbb{N}$ is not bounded.

Counterexample: Bounded does not imply compact

The interval $(0, 1)$ is bounded but not compact in $\mathbb{R}$.

Compactness: open cover definition

Definition: Compactness

A set $K \subset X$ is compact if every open cover of K admits a finite subcover. That is, if

$$K \subset \bigcup_{i \in I} U_i$$

with each U_i open, then there exist $i_1, \dots, i_m$ such that

$$K \subset U_{i_1} \cup \dots \cup U_{i_m}.$$

Interpretation

An infinite amount of local information can be reduced to finitely many pieces.

Sequential compactness

Theorem: Metric spaces

In metric spaces, compactness is equivalent to sequential compactness: every sequence in K has a subsequence converging to a point of K .

Example: Why $[0, 1]$ is compact

Every sequence in $[0, 1]$ has a convergent subsequence, and the limit remains in $[0, 1]$.

Counterexample: Why $(0, 1)$ is not compact

The sequence $x_n = 1/n$ lies in $(0, 1)$ but every convergent subsequence has limit $0 \notin (0, 1)$.

Heine-Borel in $\mathbb{R}^n$

Theorem: Heine-Borel theorem

A subset $K \subset \mathbb{R}^n$ is compact if and only if it is closed and bounded.

Important limitation

This theorem is special to finite-dimensional Euclidean spaces. In infinite-dimensional spaces, closed and bounded does not imply compact.

Example: Examples in $\mathbb{R}^2$

- Closed disk $\{x \in \mathbb{R}^2 : \|x\|_2 \leq 1\}$: compact.
- Open disk $\{x \in \mathbb{R}^2 : \|x\|_2 < 1\}$: not compact.
- Closed half-line $[0, +\infty)$: closed but not bounded, not compact.

Continuous image of compact sets

Theorem: Compactness is preserved

If K is compact and $f : K \rightarrow Y$ is continuous, then $f(K)$ is compact.

Proof idea: Open-cover idea

Take an open cover of $f(K)$. Pull it back through f to get an open cover of K . Compactness of K gives a finite subcover. Push it forward to cover $f(K)$.

Extreme value theorem

Theorem: Maximum and minimum

If K is compact and $f : K \rightarrow \mathbb{R}$ is continuous, then there exist $x_{min}, x_{max} \in K$ such that

$$f(x_{min}) \leq f(x) \leq f(x_{max}) \quad \forall x \in K.$$

Example: Polynomial on a closed interval

The function $f(x) = x^3 - 3x$ has a maximum and a minimum on $[-2, 2]$.

Counterexamples to extreme values

Counterexample: Missing compactness

- ① $f(x) = x$ on $(0, 1)$ has no maximum and no minimum.
- ② $f(x) = 1/x$ on $(0, 1)$ is continuous but unbounded.
- ③ $f(x) = x^2$ on $\mathbb{R}$ has a minimum but no maximum.

Lesson

Continuity alone is not enough. Compactness is the hypothesis that forces existence.

Total boundedness

Definition: Totally bounded

A metric space X is totally bounded if, for every $\varepsilon > 0$, there exist finitely many points $x_1, \dots, x_m$ such that

$$X \subset \bigcup_{j=1}^m B(x_j, \varepsilon).$$

Theorem: Compactness in metric spaces

A metric space is compact if and only if it is complete and totally bounded.

Interpretation

Completeness controls limits; total boundedness controls escape to infinity and excessive oscillation.

Exam-style problem: compactness

Exercise: Question

For each set, decide whether it is compact in $\mathbb{R}$:

- ① $A = [0, 1] \cup \{2\}$.
- ② $B = \{1/n : n \in \mathbb{N}\}$.
- ③ $C = \{0\} \cup \{1/n : n \in \mathbb{N}\}$.
- ④ $D = [0, +\infty)$.

Solution plan

Use Heine-Borel: compact in $\mathbb{R}$ means closed and bounded.

Solution: compactness exam problem

Answers

- ① A is closed and bounded: compact.
- ② B is bounded but not closed because 0 is a limit point not in B : not compact.
- ③ C is closed and bounded: compact.
- ④ D is closed but unbounded: not compact.

Counterexample: Most common error

Students often say that B is compact because it is bounded. Boundedness alone is not enough.

Module 1

Normed spaces

Norms: one concept, many choices

ℓ^1 length

$$\|x\|_1 = \sum_i |x_i|.$$

Adds coordinate errors.

ℓ^2 length

$$\|x\|_2 = \sqrt{\sum_i x_i^2}.$$

Euclidean geometry.

ℓ^∞ length

$$\|x\|_\infty = \max_i |x_i|.$$

Worst coordinate error.

Exercise: Compute

For $x = (2, -1, 4)$, compute $\|x\|_1$, $\|x\|_2$, $\|x\|_\infty$.

Solution: three norms of one vector

For $x = (2, -1, 4)$:

$$\|x\|_1 = 2 + 1 + 4 = 7,$$

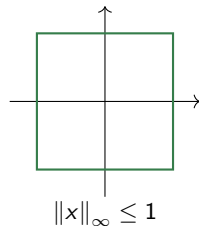
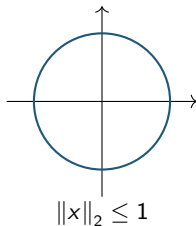
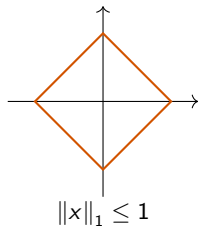
$$\|x\|_2 = \sqrt{2^2 + (-1)^2 + 4^2} = \sqrt{21},$$

$$\|x\|_\infty = \max\{2, 1, 4\} = 4.$$

Intuition: Same vector, different emphasis

$\|\cdot\|_1$ measures total size, $\|\cdot\|_2$ measures Euclidean size, $\|\cdot\|_\infty$ measures the largest single coordinate.

Unit balls for norms in $\mathbb{R}^2$



Visual memory

ℓ^1 diamond, ℓ^2 circle, ℓ^∞ square.

Exam-style: is this a norm?

Exercise: Candidate

On $\mathbb{R}^2$, define

$$p(x_1, x_2) = |x_1|.$$

Is p a norm?

Check the axioms

- 1 Non-negativity.
- 2 Identity: $p(x) = 0 \Rightarrow x = 0$?
- 3 Homogeneity.
- 4 Triangle inequality.

Solution: not a norm

Counterexample: Failure of identity

Take

$$x = (0, 1) \neq (0, 0).$$

Then

$$p(x) = |0| = 0.$$

Therefore $p(x) = 0$ does not imply $x = 0$.

Lesson

A formula may measure part of a vector, but a norm must detect every non-zero vector.

Equivalent norms: finite-dimensional comfort

Theorem: Finite dimension

On $\mathbb{R}^n$, all norms are equivalent: if $\|\cdot\|_a$ and $\|\cdot\|_b$ are norms, then there exist $c, C > 0$ such that

$$c \|x\|_a \leq \|x\|_b \leq C \|x\|_a.$$

Intuition: Meaning

The numerical value of length changes, but the notion of convergence does not.

Worked inequalities between norms

For $x \in \mathbb{R}^n$,

$$\|x\|_{\infty} \leq \|x\|_2 \leq \|x\|_1 \leq \sqrt{n} \|x\|_2 \leq n \|x\|_{\infty}.$$

Worked example: Example in $\mathbb{R}^3$

For $x = (2, -1, 4)$:

$$4 = \|x\|_{\infty} \leq \sqrt{21} = \|x\|_2 \leq 7 = \|x\|_1.$$

Exercise: Prove one part

Show that $\|x\|_2 \leq \|x\|_1$.

Why linearity is useful

Intuition: Linear maps preserve structure

A linear map sends combinations to combinations:

$$T(\lambda x + \mu y) = \lambda T(x) + \mu T(y).$$

This makes them predictable and easier to estimate.

Worked example: Matrix example

Let

$$A = \begin{pmatrix} 1 & 2 \\ 0 & 1 \end{pmatrix}, \quad x = \begin{pmatrix} x_1 \\ x_2 \end{pmatrix}.$$

Then

$$Ax = \begin{pmatrix} x_1 + 2x_2 \\ x_2 \end{pmatrix}.$$

Operator norm: worst amplification

Definition: Operator norm

For a linear map $T : X \rightarrow Y$ between normed spaces,

$$\|T\| = \sup_{\|x\| \leq 1} \|Tx\|.$$

Intuition: Practical meaning

$\|T\|$ is the largest possible output size generated by an input of size at most one.

Exercise: Compute

For $T(x) = 3x$ on $\mathbb{R}$, compute $\|T\|$.

Solution: operator norm on $\mathbb{R}$

Worked example: Scaling map

If $T(x) = 3x$, then for $|x| \leq 1$,

$$|T(x)| = |3x| = 3|x| \leq 3.$$

The value 3 is attained at $x = 1$ or $x = -1$. Therefore

$$\|T\| = 3.$$

Lesson

The operator norm is not mysterious: it is the maximal stretching factor.

Vector spaces: reminder

Definition: Vector space

A vector space over $\mathbb{R}$ is a set V with addition and scalar multiplication satisfying the usual algebraic rules.

Example: Examples

- $\mathbb{R}^n$.
- The set of real polynomials.
- The set $C([a, b])$ of continuous real-valued functions on $[a, b]$.
- The set of real sequences $(x_n)_{n \geq 1}$.

Norms: definition

Definition: Norm

Let V be a vector space. A function $\|\cdot\| : V \rightarrow [0, +\infty)$ is a norm if, for all $x, y \in V$ and $\lambda \in \mathbb{R}$,

$$\|x\| = 0 \iff x = 0,$$

$$\|\lambda x\| = |\lambda| \|x\|,$$

$$\|x + y\| \leq \|x\| + \|y\|.$$

From norm to metric

Every norm induces a metric:

$$d(x, y) = \|x - y\|.$$

Three standard norms on $\mathbb{R}^n$

Example: For $x = (x_1, \dots, x_n)$

$$\begin{aligned}\|x\|_1 &= \sum_{i=1}^n |x_i|, & \|x\|_2 &= \left(\sum_{i=1}^n |x_i|^2 \right)^{1/2}, \\ \|x\|_\infty &= \max_{1 \leq i \leq n} |x_i|.\end{aligned}$$

Exercise: Compute

For $x = (2, -1, 3)$, compute $\|x\|_1$, $\|x\|_2$ and $\|x\|_\infty$.

Solution: three norms

Computation

For $x = (2, -1, 3)$:

$$\|x\|_1 = |2| + |-1| + |3| = 6.$$

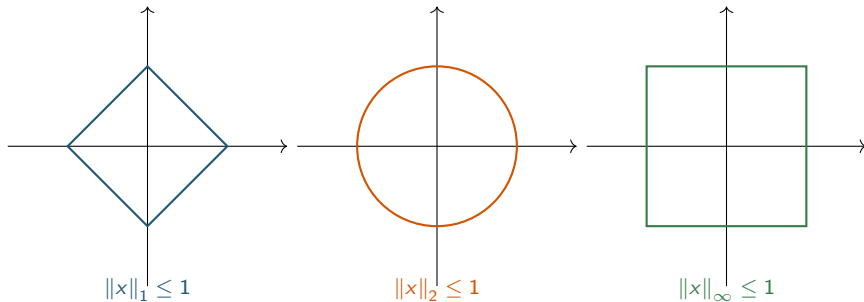
$$\|x\|_2 = \sqrt{2^2 + (-1)^2 + 3^2} = \sqrt{14}.$$

$$\|x\|_\infty = \max\{2, 1, 3\} = 3.$$

Interpretation

Different norms measure size differently, but in finite dimension they define the same notion of convergence.

Unit balls for different norms



Geometry

A norm is also a geometry: the unit ball determines how distances are perceived.

Proving that $\|\cdot\|_\infty$ is a norm

Proof idea: Triangle inequality

For $x, y \in \mathbb{R}^n$,

$$|x_i + y_i| \leq |x_i| + |y_i| \leq \|x\|_\infty + \|y\|_\infty$$

for each i . Taking the maximum over i gives

$$\|x + y\|_\infty \leq \|x\|_\infty + \|y\|_\infty.$$

The other two norm axioms are immediate.

Counterexample: not a norm

Counterexample: A tempting formula

On $\mathbb{R}^2$, define

$$p(x_1, x_2) = |x_1|.$$

Then p is nonnegative and homogeneous, and satisfies the triangle inequality. But

$$p(0, 1) = 0$$

although $(0, 1) \neq (0, 0)$.

Lesson

All norm axioms matter. This object is a seminorm, not a norm.

Equivalent norms in finite dimension

Theorem: Finite-dimensional equivalence

On $\mathbb{R}^n$, all norms are equivalent: if $\|\cdot\|_a$ and $\|\cdot\|_b$ are norms, then there exist constants $c, C > 0$ such that

$$c \|x\|_a \leq \|x\|_b \leq C \|x\|_a$$

for every $x \in \mathbb{R}^n$.

Consequence

In finite dimension, the choice of norm does not change which sequences converge.

Example: inequalities among standard norms

Example: For every $x \in \mathbb{R}^n$

$$\|x\|_{\infty} \leq \|x\|_2 \leq \|x\|_1 \leq \sqrt{n} \|x\|_2 \leq n \|x\|_{\infty}.$$

Proof idea: One inequality

Since $|x_i| \leq \|x\|_{\infty}$ for each i ,

$$\|x\|_1 = \sum_{i=1}^n |x_i| \leq n \|x\|_{\infty}.$$

Convergence in normed spaces

Definition: Norm convergence

A sequence (x_n) in a normed space $(V, \|\cdot\|)$ converges to $x \in V$ if

$$\|x_n - x\| \rightarrow 0.$$

Example: In $\mathbb{R}^2$

Let $x_n = (1/n, 2/n)$. Then

$$\|x_n\|_2 = \frac{\sqrt{5}}{n} \rightarrow 0.$$

Thus $x_n \rightarrow (0, 0)$.

Infinite dimension: norms may not be equivalent

Counterexample: On $C([0, 1])$

Consider the norms

$$\|f\|_1 = \int_0^1 |f(t)| dt, \quad \|f\|_\infty = \sup_{t \in [0,1]} |f(t)|.$$

There is no constant C such that

$$\|f\|_\infty \leq C \|f\|_1$$

for all continuous f .

Reason

A continuous spike can have very small area but very large height.

Exercise: spike counterexample

Exercise: Construct explicitly

For each n , define a triangular function f_n on $[0, 1]$ with height 1 and base length $2/n$ centered at $1/2$.

- 1 Compute $\|f_n\|_\infty$.
- 2 Estimate $\|f_n\|_1$.
- 3 Explain why this disproves equivalence of $\|\cdot\|_1$ and $\|\cdot\|_\infty$.

Solution: spike counterexample

Computation

The height is 1, hence

$$\|f_n\|_{\infty} = 1.$$

The area of the triangle is

$$\|f_n\|_1 = \int_0^1 |f_n(t)| dt = \frac{1}{2} \cdot \frac{2}{n} \cdot 1 = \frac{1}{n}.$$

If $\|f\|_{\infty} \leq C \|f\|_1$ held for all f , then

$$1 \leq C/n$$

for all n , impossible.

Module 1

Sequence spaces and inequalities

Sequence spaces as infinite tables

Intuition: From finite vectors to infinite vectors

A sequence $x = (x_1, x_2, \dots)$ is an infinite-dimensional vector. Sequence spaces decide which infinite vectors are admissible.

ℓ^1

$$\sum_{n=1}^{\infty} |x_n| < \infty.$$

Total mass is finite.

ℓ^2

$$\sum_{n=1}^{\infty} |x_n|^2 < \infty.$$

Energy is finite.

Worked example: $x_n = 1/n$

Let $x = (1, 1/2, 1/3, \dots)$.

Counterexample: Not in ℓ^1

$$\sum_{n=1}^{\infty} \frac{1}{n} = +\infty.$$

Worked example: In ℓ^2

$$\sum_{n=1}^{\infty} \frac{1}{n^2} < +\infty.$$

Thus $x \in \ell^2$.

Takeaway

$\ell^1 \subset \ell^2$, but not conversely.

Exercise: power sequences

Exercise: Classification

For $a > 0$, define

$$x_n = \frac{1}{n^a}.$$

For which values of a is $x \in \ell^1$? For which values is $x \in \ell^2$?

Recall

The series $\sum 1/n^p$ converges if and only if $p > 1$.

Solution: power sequences

Worked example: ℓ^1

$$\sum_{n=1}^{\infty} |x_n| = \sum_{n=1}^{\infty} \frac{1}{n^a}$$

converges iff $a > 1$.

Worked example: ℓ^2

$$\sum_{n=1}^{\infty} |x_n|^2 = \sum_{n=1}^{\infty} \frac{1}{n^{2a}}$$

converges iff $2a > 1$, i.e. $a > 1/2$.

Holder inequality: a usable form

Theorem: Holder

If $1 < p, q < \infty$ and

$$\frac{1}{p} + \frac{1}{q} = 1,$$

then

$$\sum_{n=1}^{\infty} |x_n y_n| \leq \|x\|_p \|y\|_q.$$

Intuition: Meaning

The product of two sequences is summable if each sequence is controlled in the matching norm.

Worked example: using Holder

Let

$$x_n = \frac{1}{n}, \quad y_n = \frac{1}{n^2}.$$

Choose $p = q = 2$. Then $x \in \ell^2$ and $y \in \ell^2$. Holder gives

$$\sum_{n=1}^{\infty} \frac{1}{n^3} = \sum_{n=1}^{\infty} |x_n y_n| \leq \|x\|_2 \|y\|_2 < \infty.$$

Exercise: Check

Could we use $p = 3$ and $q = 3/2$ instead?

Minkowski inequality: triangle inequality for ℓ^p

Theorem: Minkowski

For $1 \leq p < \infty$,

$$\|x + y\|_p \leq \|x\|_p + \|y\|_p.$$

Worked example: Finite example

Take $x = (1, 2)$, $y = (3, -1)$, $p = 2$.

$$\|x + y\|_2 = \|(4, 1)\|_2 = \sqrt{17}.$$

$$\|x\|_2 + \|y\|_2 = \sqrt{5} + \sqrt{10}.$$

Indeed $\sqrt{17} \leq \sqrt{5} + \sqrt{10}$.

Counterexample: why $p < 1$ fails

Counterexample: Not a norm

For $0 < p < 1$, the expression

$$\left(\sum |x_n|^p \right)^{1/p}$$

does not satisfy the triangle inequality.

Worked example: Example in $\mathbb{R}^2$

Take $p = 1/2$, $x = (1, 0)$, $y = (0, 1)$.

$$\|x + y\|_p = (1 + 1)^2 = 4,$$

while

$$\|x\|_p + \|y\|_p = 1 + 1 = 2.$$

Definition: ℓ^p spaces

For $1 \leq p < \infty$,

$$\ell^p = \left\{ x = (x_n)_{n \geq 1} : \sum_{n=1}^{\infty} |x_n|^p < \infty \right\},$$

with norm

$$\|x\|_p = \left(\sum_{n=1}^{\infty} |x_n|^p \right)^{1/p}.$$

Also

$$\ell^{\infty} = \{x : \sup_n |x_n| < \infty\}, \quad \|x\|_{\infty} = \sup_n |x_n|.$$

Examples in ℓ^p

Example: Power sequences

Let $x_n = 1/n^a$. Then

$$x \in \ell^p \iff \sum_{n=1}^{\infty} \frac{1}{n^{ap}} < \infty \iff ap > 1.$$

Therefore

$$x \in \ell^p \iff a > \frac{1}{p}.$$

Exercise: Classify

For $x_n = 1/n$, decide whether $x \in \ell^1, \ell^2, \ell^\infty$.

Solution: $x_n = 1/n$

Classification

$$\sum_{n=1}^{\infty} \frac{1}{n} = +\infty,$$

so $x \notin \ell^1$.

$$\sum_{n=1}^{\infty} \frac{1}{n^2} < +\infty,$$

so $x \in \ell^2$. Also

$$\sup_n \frac{1}{n} = 1,$$

so $x \in \ell^\infty$.

Holder inequality: statement

Theorem: Holder inequality

Let $1 < p, q < \infty$ with

$$\frac{1}{p} + \frac{1}{q} = 1.$$

If $x \in \ell^p$ and $y \in \ell^q$, then

$$\sum_{n=1}^{\infty} |x_n y_n| \leq \|x\|_p \|y\|_q.$$

Special case

For $p = q = 2$, Holder becomes Cauchy-Schwarz.

Young inequality behind Holder

Theorem: Young inequality

If $a, b \geq 0$ and $p, q > 1$ with $1/p + 1/q = 1$, then

$$ab \leq \frac{a^p}{p} + \frac{b^q}{q}.$$

Proof idea: Idea

Use convexity of the exponential or minimize

$$\frac{a^p}{p} + \frac{b^q}{q} - ab.$$

Equality holds when $a^p = b^q$.

Holder inequality: proof sketch

Proof idea: Normalize first

Assume $\|x\|_p = \|y\|_q = 1$. Young's inequality gives

$$|x_n y_n| \leq \frac{|x_n|^p}{p} + \frac{|y_n|^q}{q}.$$

Summing over n ,

$$\sum |x_n y_n| \leq \frac{1}{p} \sum |x_n|^p + \frac{1}{q} \sum |y_n|^q = \frac{1}{p} + \frac{1}{q} = 1.$$

Rescale to obtain the general result.

Minkowski inequality

Theorem: Minkowski inequality

For $1 \leq p < \infty$ and $x, y \in \ell^p$,

$$\|x + y\|_p \leq \|x\|_p + \|y\|_p.$$

Thus $\|\cdot\|_p$ is a norm.

Meaning

Minkowski is the triangle inequality in ℓ^p .

Example: using Cauchy-Schwarz

Example: Bounding a weighted sum

Let $x \in \ell^2$. Then

$$\sum_{n=1}^{\infty} \frac{|x_n|}{n} \leq \left(\sum_{n=1}^{\infty} |x_n|^2 \right)^{1/2} \left(\sum_{n=1}^{\infty} \frac{1}{n^2} \right)^{1/2}.$$

So the series converges absolutely.

Pattern

Pair an unknown sequence with a known summable weight.

Exam-style: Holder application

Exercise: Question

Let $x \in \ell^3$ and define

$$y_n = \frac{1}{n^{2/3}}.$$

Show that $\sum_{n=1}^{\infty} |x_n|/n^{2/3}$ converges if $y \in \ell^{3/2}$. Then determine whether $y \in \ell^{3/2}$.

Hint

Use $p = 3$ and $q = 3/2$.

Solution: Holder application

Step 1

Holder gives

$$\sum |x_n y_n| \leq \|x\|_3 \|y\|_{3/2}.$$

Step 2

Here

$$\|y\|_{3/2}^{3/2} = \sum_{n=1}^{\infty} \left(\frac{1}{n^{2/3}} \right)^{3/2} = \sum_{n=1}^{\infty} \frac{1}{n} = +\infty.$$

So $y \notin \ell^{3/2}$, and this Holder argument fails.

Counterexample: Lesson

A theorem cannot be applied if one hypothesis is missing.

Inclusion of ℓ^p spaces

Theorem: Sequence-space inclusion

For sequences, if $1 \leq p < q \leq \infty$, then

$$\ell^p \subset \ell^q.$$

Example: Example

If $x \in \ell^1$, then $x \in \ell^2$ and $x \in \ell^\infty$.

Counterexample: Not conversely

$x_n = 1/n$ belongs to ℓ^2 but not to ℓ^1 .

Proof idea: $\ell^p \subset \ell^q$

Proof idea: Key observation

If $x \in \ell^p$, then $x_n \rightarrow 0$. Hence for n large enough, $|x_n| \leq 1$. If $p < q$, then for large n ,

$$|x_n|^q \leq |x_n|^p.$$

The tail of $\sum |x_n|^q$ is dominated by the tail of $\sum |x_n|^p$, and finitely many initial terms do not matter.

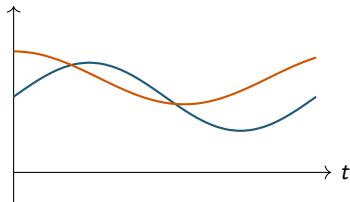
Module 1

Function spaces

Functions as points

Intuition: A function space is a space of objects

In $C[0, 1]$, each point is an entire curve. Distances compare curves.

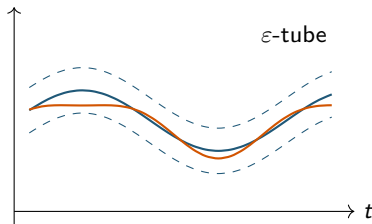


Sup distance

$$d_{\infty}(f, g) = \sup_{t \in [0, 1]} |f(t) - g(t)|.$$

It measures the largest vertical gap.

Uniform convergence: picture and meaning



Intuition: Uniform convergence

Eventually the whole graph of f_n lies inside an ε -tube around f .

Exercise: Compare

How is this stronger than pointwise convergence?

Pointwise but not uniform convergence

Counterexample: Standard example

On $[0, 1]$, let

$$f_n(x) = x^n.$$

Then $f_n(x) \rightarrow 0$ for $0 \leq x < 1$, while $f_n(1) = 1$ for all n . The pointwise limit is discontinuous:

$$f(x) = \begin{cases} 0, & 0 \leq x < 1, \\ 1, & x = 1. \end{cases}$$

Conclusion

The convergence cannot be uniform because a uniform limit of continuous functions is continuous.

Exercise: sup norm computation

Exercise: Compute a distance

Let $f(t) = t$ and $g(t) = t^2$ on $[0, 1]$. Compute

$$\|f - g\|_{\infty}.$$

Hint

Maximize

$$h(t) = t - t^2 = t(1 - t)$$

on $[0, 1]$.

Solution: distance between t and t^2

Worked example: Maximize the vertical gap

For $t \in [0, 1]$,

$$|t - t^2| = t - t^2.$$

The maximum occurs when

$$h'(t) = 1 - 2t = 0 \quad \Rightarrow \quad t = \frac{1}{2}.$$

Thus

$$\|f - g\|_{\infty} = h(1/2) = \frac{1}{4}.$$

Completeness of $C[a, b]$ with the sup norm

Theorem: Key fact

The space $C[a, b]$ with the norm

$$\|f\|_{\infty} = \sup_{t \in [a, b]} |f(t)|$$

is complete.

Intuition: Why this matters

If continuous functions f_n stabilize uniformly, their limit is still continuous. The space does not lose the limit.

Polynomial approximation as a distance problem

Intuition: Approximation problem

Given a function f , find a simpler function p such that

$$\|f - p\|$$

is small.

Worked example: Example

Approximate $f(t) = e^t$ on $[0, 1]$ by the first-order Taylor polynomial

$$p(t) = 1 + t.$$

The error is measured by $\|f - p\|_{\infty}$.

Exercise: Task

Use Taylor's theorem to give an upper bound for the error.

The space $C([a, b])$

Definition: Continuous functions

$$C([a, b]) = \{f : [a, b] \rightarrow \mathbb{R} : f \text{ is continuous}\}.$$

With the sup norm

$$\|f\|_{\infty} = \sup_{t \in [a, b]} |f(t)|,$$

this is a normed space.

Why the supremum exists

A continuous function on the compact interval $[a, b]$ is bounded and attains its maximum absolute value.

Uniform convergence

Definition: Uniform convergence

A sequence $f_n \in C([a, b])$ converges uniformly to f if

$$\|f_n - f\|_{\infty} = \sup_{t \in [a, b]} |f_n(t) - f(t)| \rightarrow 0.$$

Meaning

The same error bound works for all points t in the interval.

Pointwise versus uniform convergence

Counterexample: A standard sequence

Let $f_n(x) = x^n$ on $[0, 1]$. Then

$$f_n(x) \rightarrow f(x) = \begin{cases} 0, & 0 \leq x < 1, \\ 1, & x = 1. \end{cases}$$

The convergence is pointwise but not uniform.

Reason

The pointwise limit is not continuous, while a uniform limit of continuous functions must be continuous.

Uniform limit of continuous functions

Theorem: Stability of continuity

If $f_n \in C([a, b])$ and $f_n \rightarrow f$ uniformly, then $f \in C([a, b])$.

Proof idea: $\varepsilon/3$ argument

Fix x_0 . Choose n such that $\|f_n - f\|_\infty < \varepsilon/3$. By continuity of f_n , choose δ such that

$$|x - x_0| < \delta \Rightarrow |f_n(x) - f_n(x_0)| < \varepsilon/3.$$

Then add and subtract $f_n(x)$ and $f_n(x_0)$.

$C([a, b])$ is complete under the sup norm

Theorem: Completeness

The normed space $(C([a, b]), \|\cdot\|_\infty)$ is complete.

Proof idea: Idea

If (f_n) is Cauchy in sup norm, then for each t , $(f_n(t))$ is Cauchy in $\mathbb{R}$, hence converges to some $f(t)$. The Cauchy property is uniform, so $f_n \rightarrow f$ uniformly. Since uniform limits of continuous functions are continuous, $f \in C([a, b])$.

An incomplete function space

Counterexample: Polynomials under sup norm

Let P be the space of polynomials on $[0, 1]$ with $\|\cdot\|_\infty$. It is not complete.

Reason

Polynomials can converge uniformly to non-polynomial continuous functions, for example to e^x through Taylor polynomials. The limit e^x is continuous but not a polynomial, so it is outside P .

Spaces $C^k([a, b])$

Definition: Smoothness spaces

For $k \in \mathbb{N}$,

$$C^k([a, b]) = \{f : f, f', \dots, f^{(k)} \text{ are continuous on } [a, b]\}.$$

A natural norm is

$$\|f\|_{C^k} = \sum_{j=0}^k \|f^{(j)}\|_{\infty}.$$

Example: For $k = 1$

$$\|f\|_{C^1} = \|f\|_{\infty} + \|f'\|_{\infty}.$$

Derivative as a linear operator

Example: Derivative map

Define

$$D : C^1([a, b]) \rightarrow C([a, b]), \quad Df = f'.$$

Then D is linear:

$$D(\lambda f + g) = \lambda Df + Dg.$$

Moreover, with the C^1 norm,

$$\|Df\|_{\infty} = \|f'\|_{\infty} \leq \|f\|_{C^1},$$

so D is bounded.

Derivative is not bounded on $C([a, b])$

Counterexample: Wrong domain

If one tries to view differentiation as

$$D : C([0, 1]) \rightarrow C([0, 1]),$$

it is not even defined on every continuous function.

Counterexample: Even on polynomials with sup norm

Let $f_n(x) = x^n$ on $[0, 1]$. Then

$$\|f_n\|_{\infty} = 1, \quad \|f'_n\|_{\infty} = n.$$

No uniform bound $\|Df\| \leq C \|f\|$ can hold on polynomials with the sup norm.

Hölder spaces: motivation

Regularity between continuity and differentiability

A function may be continuous but too irregular to be differentiable. Hölder continuity measures intermediate smoothness.

Definition: Hölder condition

For $\gamma \in (0, 1]$, a function $f : [a, b] \rightarrow \mathbb{R}$ is γ -Hölder if there exists $C > 0$ such that

$$|f(x) - f(y)| \leq C|x - y|^\gamma$$

for all $x, y \in [a, b]$.

Examples of Hölder functions

Example: Square root

On $[0, 1]$, $f(x) = \sqrt{x}$ is $1/2$ -Hölder:

$$|\sqrt{x} - \sqrt{y}| \leq \sqrt{|x - y|}.$$

It is not Lipschitz on $[0, 1]$.

Example: Lipschitz means Hölder with exponent 1

If $|f(x) - f(y)| \leq L|x - y|$, then f is 1-Hölder.

Hölder seminorm

Definition: Hölder seminorm

$$[f]_\gamma = \sup_{x \neq y} \frac{|f(x) - f(y)|}{|x - y|^\gamma}.$$

A norm on the Hölder space is

$$\|f\|_{C^{0,\gamma}} = \|f\|_\infty + [f]_\gamma.$$

Why seminorm?

Constants have $[f]_\gamma = 0$ but are not the zero function. Adding $\|f\|_\infty$ gives a norm.

Exam-style: Hölder regularity

Exercise: Question

Show that $f(x) = |x|^\gamma$ on $[-1, 1]$ is γ -Hölder for $0 < \gamma \leq 1$. Is it Lipschitz when $0 < \gamma < 1$?

Answer sketch

Use the inequality

$$||x|^\gamma - |y|^\gamma| \leq |x - y|^\gamma.$$

For $0 < \gamma < 1$, the quotient

$$\frac{|x|^\gamma}{|x|} = |x|^{\gamma-1}$$

blows up as $x \rightarrow 0$, so it is not Lipschitz.

Module 1

Banach spaces and linear operators

Banach space: a safe working environment

Intuition: Core idea

A Banach space is a normed space where Cauchy approximations have limits inside the space.

Good situation

Algorithm produces a Cauchy sequence. Completeness guarantees a candidate limit in the space.

Counterexample: Bad situation

The approximations stabilize, but the limit lives outside the chosen space.

Bounded linear maps: the simple test

Theorem: Equivalent conditions

For a linear map $T : X \rightarrow Y$ between normed spaces, the following are equivalent:

- ① T is continuous.
- ② T is continuous at 0.
- ③ There exists $C > 0$ such that $\|Tx\| \leq C \|x\|$ for all x .

Practical rule

To prove continuity of a linear map, find a constant C .

Worked example: an integral operator

Worked example: Operator on $C[0, 1]$

Define

$$(Tf)(x) = \int_0^x f(t) dt.$$

Then

$$|Tf(x)| \leq \int_0^x |f(t)| dt \leq \|f\|_{\infty}.$$

Taking the supremum over x ,

$$\|Tf\|_{\infty} \leq \|f\|_{\infty}.$$

Therefore T is bounded and $\|T\| \leq 1$.

Exercise: exact norm of the integral operator

Exercise: Find $\|T\|$

For

$$(Tf)(x) = \int_0^x f(t) dt$$

on $C[0, 1]$ with the sup norm, show that

$$\|T\| = 1.$$

Hint

Use the constant function $f(t) = 1$.

Solution: exact operator norm

We already know $\|T\| \leq 1$. For $f(t) = 1$,

$$\|f\|_{\infty} = 1,$$

and

$$(Tf)(x) = \int_0^x 1 \, dt = x.$$

Hence

$$\|Tf\|_{\infty} = 1.$$

Therefore

$$\|T\| = 1.$$

Kernel and range: two basic subspaces

Definition: Kernel and range

For a linear map $T : X \rightarrow Y$,

$$\text{Ker}(T) = \{x \in X : T(x) = 0\},$$

$$\text{Range}(T) = \{T(x) : x \in X\}.$$

Worked example: Matrix example

For

$$A = \begin{pmatrix} 1 & 1 \\ 2 & 2 \end{pmatrix},$$

$$\text{Ker}(A) = \{(t, -t) : t \in \mathbb{R}\}.$$

Exercise: kernel and range

Exercise: Compute

Let

$$A = \begin{pmatrix} 1 & 2 & 0 \\ 0 & 1 & 1 \end{pmatrix}.$$

Find:

- ① $\text{Ker}(A)$;
- ② $\text{Range}(A)$;
- ③ the rank of A .

Hint

Solve $Ax = 0$ and identify the span of the columns.

Closedness matters for approximation

Theorem: Best approximation idea

If a set is closed and the ambient space is complete, minimizing sequences have a chance to converge to a feasible point.

Counterexample: If the set is not closed

The infimum distance can be approached but not attained. Example in $\mathbb{R}$:

$$A = (0, 1), \quad x = 2.$$

The closest distance is 1, but no point of A attains it.

Banach spaces

Definition: Banach space

A Banach space is a normed vector space that is complete with respect to the metric induced by its norm.

Example: Examples

- $\mathbb{R}^n$ with any norm.
- $C([a, b])$ with $\|\cdot\|_\infty$.
- ℓ^p for $1 \leq p \leq \infty$.
- $C^k([a, b])$ with the C^k norm.

Why Banach spaces matter

Completeness as a guarantee

If a sequence of approximations is Cauchy in a Banach space, then it converges to an element of the same space.

Example: Typical use

- Build approximate solutions (x_n) .
- Prove (x_n) is Cauchy using an estimate.
- Use completeness to obtain a limit x .
- Prove x solves the original problem.

Linear maps

Definition: Linear operator

Let X, Y be vector spaces. A map $T : X \rightarrow Y$ is linear if

$$T(\lambda x + z) = \lambda T(x) + T(z)$$

for every $x, z \in X$ and scalar λ .

Example: Examples

- Matrix multiplication $T(x) = Ax$.
- Differentiation $Df = f'$ on a suitable domain.
- Integration operator $(Tf)(x) = \int_a^x f(t) dt$.

Kernel and range

Definition: Kernel and range

For a linear map $T : X \rightarrow Y$,

$$\text{Ker}(T) = \{x \in X : T(x) = 0\},$$

$$\text{Range}(T) = \{T(x) : x \in X\} \subset Y.$$

Example: Derivative

For $D : C^1([a, b]) \rightarrow C([a, b])$, $Df = f'$:

$$\text{Ker}(D) = \{\text{constant functions}\}.$$

Bounded linear maps

Definition: Bounded operator

Let X, Y be normed spaces. A linear map $T : X \rightarrow Y$ is bounded if there exists $C \geq 0$ such that

$$\|Tx\|_Y \leq C \|x\|_X$$

for all $x \in X$.

Operator norm

$$\|T\| = \sup_{x \neq 0} \frac{\|Tx\|_Y}{\|x\|_X} = \sup_{\|x\|_X \leq 1} \|Tx\|_Y.$$

Bounded iff continuous

Theorem: Linear maps between normed spaces

For a linear map $T : X \rightarrow Y$, the following are equivalent:

- ① T is continuous on X .
- ② T is continuous at 0 .
- ③ T is bounded.

Key message

For linear maps, continuity is equivalent to a global norm estimate.

Proof: bounded implies continuous

Proof idea: Direct estimate

Assume $\|Tx\| \leq C\|x\|$. Then

$$\|Tx - Ty\| = \|T(x - y)\| \leq C\|x - y\|.$$

Thus T is Lipschitz, hence continuous.

Proof: continuity at zero implies boundedness

Proof idea: Contradiction argument

If T is not bounded, for each n there exists x_n such that

$$\|Tx_n\| > n \|x_n\|.$$

Set

$$y_n = \frac{x_n}{n \|x_n\|}.$$

Then $\|y_n\| = 1/n \rightarrow 0$, but

$$\|Ty_n\| = \frac{\|Tx_n\|}{n \|x_n\|} > 1.$$

So $Ty_n \not\rightarrow 0$, contradicting continuity at 0.

Operator norm: matrix example

Example: Diagonal matrix

Let

$$A = \begin{pmatrix} 2 & 0 \\ 0 & -3 \end{pmatrix}$$

acting on $\mathbb{R}^2$ with Euclidean norm. Then

$$\|A\|_2 = 3.$$

Reason

$$\|Ax\|_2^2 = 4x_1^2 + 9x_2^2 \leq 9(x_1^2 + x_2^2) = 9\|x\|_2^2.$$

Equality holds for $x = (0, 1)$.

Integration operator is bounded

Example: Volterra-type operator

Define $T : C([0, 1]) \rightarrow C([0, 1])$ by

$$(Tf)(x) = \int_0^x f(t) dt.$$

Then

$$|(Tf)(x)| \leq \int_0^x |f(t)| dt \leq \|f\|_{\infty}.$$

Therefore

$$\|Tf\|_{\infty} \leq \|f\|_{\infty},$$

so T is bounded and $\|T\| \leq 1$.

Finding the exact norm of the integration operator

Exercise: Question

For

$$(Tf)(x) = \int_0^x f(t)dt$$

on $C([0, 1])$ with the sup norm, prove that $\|T\| = 1$.

Solution

We already proved $\|T\| \leq 1$. Take $f \equiv 1$. Then $\|f\|_\infty = 1$ and

$$(Tf)(x) = x, \quad \|Tf\|_\infty = 1.$$

Thus $\|T\| \geq 1$, hence $\|T\| = 1$.

Unbounded linear operator example

Counterexample: Derivative on polynomials

Let P be polynomials on $[0, 1]$ with $\|\cdot\|_\infty$. Define $D : P \rightarrow P$ by $Df = f'$. Take $f_n(x) = x^n$. Then

$$\|f_n\|_\infty = 1, \quad \|Df_n\|_\infty = n.$$

No constant C can satisfy $\|Df\| \leq C \|f\|$.

Lesson

Linearity does not imply boundedness in infinite-dimensional spaces.

Finite-dimensional contrast

Theorem: Finite dimension

Every linear map between finite-dimensional normed spaces is bounded.

Meaning

The pathologies of unbounded linear operators are genuinely infinite-dimensional.

Example: Matrix map

For $A \in \mathbb{R}^{m \times n}$,

$$\|Ax\| \leq C \|x\|$$

for some C , because each component of Ax is a finite linear combination of components of x .

Exam-style: boundedness of a functional

Exercise: Question

Define $L : C([0, 1]) \rightarrow \mathbb{R}$ by

$$L(f) = f(0).$$

Show that L is linear and bounded, and compute $\|L\|$.

Solution

Linearity is immediate. Also

$$|L(f)| = |f(0)| \leq \|f\|_{\infty},$$

so $\|L\| \leq 1$. Taking $f \equiv 1$, $|L(f)| = 1$ and $\|f\|_{\infty} = 1$, hence $\|L\| = 1$.

Neumann series: inverse by approximation

Theorem: Neumann series

Let X be a Banach space and let $T : X \rightarrow X$ be bounded with $\|T\| < 1$. Then $I - T$ is invertible and

$$(I - T)^{-1} = \sum_{n=0}^{\infty} T^n.$$

Why Banach?

The series of operators converges because $\sum \|T\|^n < \infty$ and the space of bounded operators on a Banach space is complete.

Practical meaning of Neumann series

Example: Solving a linear equation

To solve

$$x - Tx = y,$$

write formally

$$x = (I - T)^{-1}y = \sum_{n=0}^{\infty} T^n y.$$

The partial sums

$$y + Ty + T^2y + \cdots + T^N y$$

are computable approximations.

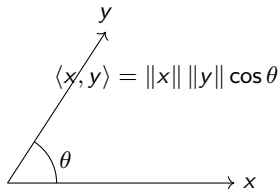
Error estimate

$$\left\| \sum_{n=N+1}^{\infty} T^n y \right\| \leq \frac{\|T\|^{N+1}}{1 - \|T\|} \|y\|.$$

Module 1

Inner product and Hilbert spaces

Inner product: length plus angle



Intuition: Geometric role

The inner product measures alignment. Positive means same general direction; zero means orthogonal; negative means opposite direction.

Worked example: dot product and angle

Let $x = (1, 2)$ and $y = (2, 1)$.

$$\langle x, y \rangle = 1 \cdot 2 + 2 \cdot 1 = 4.$$

$$\|x\| = \sqrt{5}, \quad \|y\| = \sqrt{5}.$$

Thus

$$\cos \theta = \frac{\langle x, y \rangle}{\|x\| \|y\|} = \frac{4}{5}.$$

Exercise: Compute

Find the angle between $x = (1, 0)$ and $y = (1, 1)$.

Orthogonality: cancelling information

Definition: Orthogonality

In an inner product space, x and y are orthogonal if

$$\langle x, y \rangle = 0.$$

We write $x \perp y$.

Worked example: Example

In $\mathbb{R}^3$,

$$(1, 1, 0) \perp (1, -1, 2)$$

because

$$1 \cdot 1 + 1 \cdot (-1) + 0 \cdot 2 = 0.$$

Pythagoras in inner product spaces

Theorem: Pythagorean theorem

If $x \perp y$, then

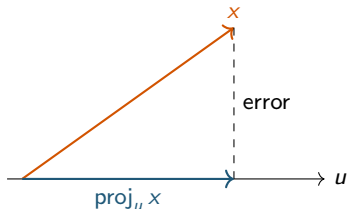
$$\|x + y\|^2 = \|x\|^2 + \|y\|^2.$$

Proof idea: One-line proof

$$\|x + y\|^2 = \langle x + y, x + y \rangle = \|x\|^2 + 2 \langle x, y \rangle + \|y\|^2.$$

If $\langle x, y \rangle = 0$, the middle term disappears.

Projection onto a line: visual formula



Definition: Projection formula

If $u \neq 0$, then

$$\text{proj}_u x = \frac{\langle x, u \rangle}{\langle u, u \rangle} u.$$

The residual $x - \text{proj}_u x$ is orthogonal to u .

Worked example: projection in $\mathbb{R}^2$

Let $x = (2, 3)$ and $u = (1, 1)$.

$$\langle x, u \rangle = 5, \quad \langle u, u \rangle = 2.$$

Therefore

$$\text{proj}_u x = \frac{5}{2}(1, 1) = \left(\frac{5}{2}, \frac{5}{2}\right).$$

The residual is

$$r = x - \text{proj}_u x = \left(-\frac{1}{2}, \frac{1}{2}\right).$$

Check:

$$\langle r, u \rangle = -\frac{1}{2} + \frac{1}{2} = 0.$$

Least squares as projection

Intuition: Approximation by a subspace

Given x and a subspace M , the best approximation $m^* \in M$ satisfies

$$x - m^* \perp M.$$

Worked example: Constant approximation

For $x = (2.1, 1.9, 2.3, 2.0)$, approximate by $c(1, 1, 1, 1)$. The projection coefficient is the average:

$$c = \frac{2.1 + 1.9 + 2.3 + 2.0}{4} = 2.075.$$

Exercise: best constant approximation

Exercise: Compute the projection

Find the best constant approximation to

$$x = (4, 1, 3, 2, 5)$$

in the Euclidean norm. That is, find c minimizing

$$\sum_{i=1}^5 (x_i - c)^2.$$

Hint

Project x onto the line spanned by $(1, 1, 1, 1, 1)$.

Solution: best constant approximation

The best constant is the average:

$$c = \frac{4 + 1 + 3 + 2 + 5}{5} = 3.$$

So the projection is

$$p = (3, 3, 3, 3, 3).$$

The residual is

$$r = x - p = (1, -2, 0, -1, 2).$$

Check orthogonality:

$$\langle r, (1, 1, 1, 1, 1) \rangle = 1 - 2 + 0 - 1 + 2 = 0.$$

Cauchy-Schwarz: the fundamental inequality

Theorem: Cauchy-Schwarz

In an inner product space,

$$|\langle x, y \rangle| \leq \|x\| \|y\|.$$

Intuition: Geometric meaning

The absolute value of the alignment cannot exceed the product of the lengths.

Exercise: Check by numbers

Verify the inequality for $x = (1, 2, 3)$ and $y = (3, 0, 1)$.

Solution: Cauchy-Schwarz by numbers

For $x = (1, 2, 3)$ and $y = (3, 0, 1)$:

$$\langle x, y \rangle = 1 \cdot 3 + 2 \cdot 0 + 3 \cdot 1 = 6.$$

$$\|x\| = \sqrt{1 + 4 + 9} = \sqrt{14}, \quad \|y\| = \sqrt{9 + 0 + 1} = \sqrt{10}.$$

Thus

$$|\langle x, y \rangle| = 6 \leq \sqrt{140} \approx 11.83.$$

Orthonormal systems: coordinates by projection

Definition: Orthonormal system

A family (e_i) is orthonormal if

$$\langle e_i, e_j \rangle = 0 \quad (i \neq j), \quad \|e_i\| = 1.$$

Coordinate formula

If $(e_1, \dots, e_n)$ is an orthonormal basis, then

$$x = \sum_{i=1}^n \langle x, e_i \rangle e_i.$$

Bessel inequality: energy cannot increase

Theorem: Bessel inequality

If (e_i) is an orthonormal system in a Hilbert space, then

$$\sum_i |\langle x, e_i \rangle|^2 \leq \|x\|^2.$$

Intuition: Meaning

The energy captured by orthogonal coordinates cannot exceed the total energy of the vector.

Hilbert projection theorem: intuition

Theorem: Projection principle

If M is a closed convex set in a Hilbert space and x is a point, there exists a unique point $m^* \in M$ closest to x .

Intuition: Geometric picture

The closest point is where the error vector becomes perpendicular to all admissible directions.

Exercise: Concept check

Why is closedness needed? Think of the interval $(0, 1)$ and the point 2.

Inner products

Definition: Real inner product

An inner product on a real vector space H is a map $\langle \cdot, \cdot \rangle : H \times H \rightarrow \mathbb{R}$ such that

- ① $\langle x, x \rangle \geq 0$ and $\langle x, x \rangle = 0 \iff x = 0$;
- ② $\langle x, y \rangle = \langle y, x \rangle$;
- ③ $\langle \lambda x + z, y \rangle = \lambda \langle x, y \rangle + \langle z, y \rangle$.

Induced norm

$$\|x\| = \sqrt{\langle x, x \rangle}.$$

Examples of inner products

Example: Euclidean space

On $\mathbb{R}^n$,

$$\langle x, y \rangle = \sum_{i=1}^n x_i y_i.$$

Example: Continuous functions

On $C([a, b])$,

$$\langle f, g \rangle = \int_a^b f(t)g(t)dt$$

is an inner product. The induced norm is

$$\|f\|_2 = \left(\int_a^b |f(t)|^2 dt \right)^{1/2}.$$

Cauchy-Schwarz inequality

Theorem: Cauchy-Schwarz

In an inner product space,

$$|\langle x, y \rangle| \leq \|x\| \|y\|.$$

Equality holds if and only if x and y are linearly dependent.

This is the central inequality of Hilbert space geometry.

It controls angles, projections and least-squares approximations.

Proof of Cauchy-Schwarz

Proof idea: Quadratic polynomial

If $y = 0$, trivial. Otherwise, for every $\lambda \in \mathbb{R}$,

$$0 \leq \|x - \lambda y\|^2 = \|x\|^2 - 2\lambda \langle x, y \rangle + \lambda^2 \|y\|^2.$$

This quadratic in λ is always nonnegative, so its discriminant is nonpositive:

$$4 \langle x, y \rangle^2 - 4 \|x\|^2 \|y\|^2 \leq 0.$$

Thus $|\langle x, y \rangle| \leq \|x\| \|y\|$.

Triangle inequality from Cauchy-Schwarz

Proof idea: Norm induced by inner product

$$\|x + y\|^2 = \|x\|^2 + 2 \langle x, y \rangle + \|y\|^2.$$

By Cauchy-Schwarz,

$$2 \langle x, y \rangle \leq 2 \|x\| \|y\|.$$

Therefore

$$\|x + y\|^2 \leq (\|x\| + \|y\|)^2.$$

Taking square roots gives

$$\|x + y\| \leq \|x\| + \|y\|.$$

Angles and orthogonality

Definition: Orthogonality

Two vectors x, y in an inner product space are orthogonal if

$$\langle x, y \rangle = 0.$$

We write $x \perp y$.

Example: In $\mathbb{R}^2$

$(1, 2) \perp (2, -1)$ because

$$1 \cdot 2 + 2 \cdot (-1) = 0.$$

Pythagoras theorem

Theorem: Pythagoras

If $x \perp y$, then

$$\|x + y\|^2 = \|x\|^2 + \|y\|^2.$$

Proof idea: One line

$$\|x + y\|^2 = \langle x + y, x + y \rangle = \|x\|^2 + 2 \langle x, y \rangle + \|y\|^2.$$

Since $\langle x, y \rangle = 0$, the result follows.

Parallelogram identity

Theorem: Parallelogram law

In every inner product space,

$$\|x + y\|^2 + \|x - y\|^2 = 2 \|x\|^2 + 2 \|y\|^2 .$$

Counterexample: Test for non-Hilbert norms

If a norm does not satisfy the parallelogram identity, it cannot come from an inner product.

ℓ^p is Hilbert only for $p = 2$

Counterexample: ℓ^1 example in $\mathbb{R}^2$

Use the ℓ^1 norm on $\mathbb{R}^2$. Let $x = (1, 0)$ and $y = (0, 1)$. Then

$$\|x + y\|_1^2 + \|x - y\|_1^2 = 2^2 + 2^2 = 8.$$

But

$$2\|x\|_1^2 + 2\|y\|_1^2 = 2 + 2 = 4.$$

The parallelogram law fails. Therefore $\|\cdot\|_1$ is not induced by an inner product.

Hilbert spaces

Definition: Hilbert space

A Hilbert space is an inner product space that is complete with respect to the norm induced by the inner product.

Example: Examples

- $\mathbb{R}^n$ with the Euclidean inner product.
- ℓ^2 with $\langle x, y \rangle = \sum_{n=1}^{\infty} x_n y_n$.
- L^2 spaces after identifying functions equal almost everywhere.

$C([0, 1])$ with L^2 norm is not complete

Counterexample: Incomplete Hilbert-like space

On $C([0, 1])$, define

$$\|f\|_2 = \left(\int_0^1 |f(t)|^2 dt \right)^{1/2}.$$

This norm comes from an inner product, but $C([0, 1])$ is not complete under it.

Reason

Continuous functions can converge in L^2 to a discontinuous function, for example approximations of a step function.

Orthogonal families

Definition: Orthogonal and orthonormal

A family $(e_i)_{i \in I}$ is orthogonal if

$$\langle e_i, e_j \rangle = 0 \quad \text{for } i \neq j.$$

It is orthonormal if additionally

$$\|e_i\| = 1 \quad \text{for all } i.$$

Example: Standard basis

In $\mathbb{R}^n$, the vectors $e_1 = (1, 0, \dots, 0), \dots, e_n = (0, \dots, 1)$ form an orthonormal basis.

Coordinates in an orthonormal basis

Theorem: Finite-dimensional formula

If $(e_1, \dots, e_n)$ is an orthonormal basis of $\mathbb{R}^n$, then

$$x = \sum_{j=1}^n \langle x, e_j \rangle e_j.$$

Moreover,

$$\|x\|^2 = \sum_{j=1}^n |\langle x, e_j \rangle|^2.$$

Interpretation

Inner products with basis vectors are coordinates.

Gram-Schmidt process

Goal

Turn linearly independent vectors $(v_1, \dots, v_n)$ into an orthonormal family $(e_1, \dots, e_n)$ spanning the same subspace.

$$\begin{aligned} u_1 &= v_1, & e_1 &= \frac{u_1}{\|u_1\|}, \\ u_k &= v_k - \sum_{j=1}^{k-1} \langle v_k, e_j \rangle e_j, & e_k &= \frac{u_k}{\|u_k\|}. \end{aligned}$$

Meaning

At each step remove the part already explained by the previous orthonormal directions.

Example: Gram-Schmidt in $\mathbb{R}^2$

Example: Start with $v_1 = (1, 1)$, $v_2 = (1, 0)$

First

$$e_1 = \frac{1}{\sqrt{2}}(1, 1).$$

Then

$$u_2 = v_2 - \langle v_2, e_1 \rangle e_1 = (1, 0) - \frac{1}{\sqrt{2}} \frac{1}{\sqrt{2}}(1, 1) = \left(\frac{1}{2}, -\frac{1}{2}\right).$$

Thus

$$e_2 = \frac{1}{\sqrt{2}}(1, -1).$$

Projection onto a line

Theorem: Projection formula

Let $u \neq 0$ in an inner product space. The projection of x onto the line $\text{span}\{u\}$ is

$$P_u x = \frac{\langle x, u \rangle}{\|u\|^2} u.$$

Orthogonality condition

The residual

$$x - P_u x$$

is orthogonal to u .

Example: projection onto a line

Example: In $\mathbb{R}^2$

Project $x = (3, 1)$ onto the line generated by $u = (1, 1)$.

$$\langle x, u \rangle = 4, \quad \|u\|^2 = 2.$$

Thus

$$P_u x = \frac{4}{2}(1, 1) = (2, 2).$$

The residual is

$$x - P_u x = (1, -1),$$

and indeed $(1, -1) \perp (1, 1)$.

Projection onto a subspace

Theorem: Orthonormal basis formula

Let M be a finite-dimensional subspace with orthonormal basis $e_1, \dots, e_m$. Then

$$P_M x = \sum_{j=1}^m \langle x, e_j \rangle e_j.$$

Best approximation

$P_M x$ is the unique point of M minimizing

$$\|x - y\|$$

over all $y \in M$.

Least squares as projection

Example: Approximation by a line

Given data points (t_i, z_i) , approximate z by $a + bt$. This means projecting the vector

$$z = (z_1, \dots, z_n)$$

onto the subspace spanned by

$$(1, \dots, 1), \quad (t_1, \dots, t_n).$$

Normal equations

The residual is orthogonal to every vector in the approximation subspace.

Hilbert projection theorem

Theorem: Projection onto a closed convex set

Let H be a Hilbert space and let $C \subset H$ be nonempty, closed and convex. For every $x \in H$ there exists a unique $y \in C$ such that

$$\|x - y\| = \inf_{z \in C} \|x - z\|.$$

Special case

If $C = M$ is a closed subspace, then $y = P_M x$ and $x - P_M x \perp M$.

Why closedness is necessary

Counterexample: Open subspace-like set

In $\mathbb{R}$, let $C = (0, 1)$ and take $x = 2$. Then

$$\inf_{z \in C} |2 - z| = 1,$$

but no $z \in (0, 1)$ satisfies $|2 - z| = 1$.

Lesson

Without closedness, the best approximating point may lie on the boundary but outside the set.

Projection theorem: proof idea

Proof idea: For closed subspaces

Let (y_n) be a minimizing sequence in M :

$$\|x - y_n\| \rightarrow \inf_{y \in M} \|x - y\|.$$

Using the parallelogram identity, show (y_n) is Cauchy. Since M is closed in the Hilbert space, M is complete; hence $y_n \rightarrow y \in M$. This y attains the minimum. Orthogonality follows by considering $y + tz$ for $z \in M$.

Orthogonal complement

Definition: Orthogonal complement

For a subset $M \subset H$,

$$M^\perp = \{x \in H : \langle x, m \rangle = 0 \text{ for all } m \in M\}.$$

Theorem: Closed subspace decomposition

If M is a closed subspace of a Hilbert space H , then

$$H = M \oplus M^\perp.$$

Every $x \in H$ can be written uniquely as

$$x = P_M x + (x - P_M x).$$

Bessel inequality

Theorem: Bessel inequality

Let (e_n) be an orthonormal sequence in a Hilbert space H . Then, for every $x \in H$,

$$\sum_{n=1}^{\infty} |\langle x, e_n \rangle|^2 \leq \|x\|^2.$$

Meaning

The squared coordinates of x along orthonormal directions cannot contain more energy than x itself.

Proof of Bessel inequality

Proof idea: Finite partial sums

Let

$$s_N = \sum_{n=1}^N \langle x, e_n \rangle e_n.$$

Then $x - s_N \perp s_N$. By Pythagoras,

$$\|x\|^2 = \|s_N\|^2 + \|x - s_N\|^2 \geq \|s_N\|^2.$$

But

$$\|s_N\|^2 = \sum_{n=1}^N |\langle x, e_n \rangle|^2.$$

Let $N \rightarrow \infty$.

Parseval identity

Theorem: Complete orthonormal systems

If (e_n) is a complete orthonormal system in H , then for every $x \in H$,

$$x = \sum_{n=1}^{\infty} \langle x, e_n \rangle e_n$$

with convergence in norm, and

$$\|x\|^2 = \sum_{n=1}^{\infty} |\langle x, e_n \rangle|^2.$$

Bessel versus Parseval

Bessel always gives $\leq$. Parseval requires completeness and gives equality.

Fourier series as Hilbert coordinates

Example: Trigonometric basis

In $L^2([-\pi, \pi])$, functions

$$1, \quad \cos(nx), \quad \sin(nx), \quad n \geq 1,$$

form an orthogonal system after normalization. Fourier coefficients are inner products with these basis functions.

Interpretation

Fourier series is coordinate expansion in a Hilbert space.

Exam-style: projection in $\mathbb{R}^3$

Exercise: Question

Let $M = \text{span}\{(1, 1, 0), (1, 0, 1)\} \subset \mathbb{R}^3$ and let $x = (1, 2, 3)$. Find the projection of x onto M .

Solution plan

Write

$$y = a(1, 1, 0) + b(1, 0, 1).$$

Impose orthogonality:

$$x - y \perp (1, 1, 0), \quad x - y \perp (1, 0, 1).$$

Solve for a, b .

Solution: projection in $\mathbb{R}^3$

Computation

Let $u = (1, 1, 0)$ and $v = (1, 0, 1)$. Then

$$y = au + bv = (a + b, a, b).$$

Orthogonality gives

$$\langle x - y, u \rangle = 0, \quad \langle x - y, v \rangle = 0.$$

Thus

$$(1 - a - b) + (2 - a) = 0,$$

$$(1 - a - b) + (3 - b) = 0.$$

So

$$2a + b = 3, \quad a + 2b = 4.$$

Hence $a = 2/3$, $b = 5/3$, and

$$P_M x = \left(\frac{7}{3}, \frac{2}{3}, \frac{5}{3} \right).$$

Module 1

Counterexamples and exam training

Exam training: how to write a clean solution

Template

- ① Name the space and the metric/norm.
- ② State the definition or theorem.
- ③ Check each hypothesis explicitly.
- ④ Perform the computation.
- ⑤ Finish with a clear conclusion.

Counterexample: Common loss of points

Writing only the final answer without explaining why the hypotheses hold.

Exam exercise 1: metric

Exercise: Question

On $\mathbb{R}$, define

$$d(x, y) = |\arctan x - \arctan y|.$$

Is d a metric?

Hint

Use that $\arctan$ is injective and that $|a - b|$ is a metric on $\mathbb{R}$.

Solution: arctangent metric

Worked example: Answer

Yes. Let $\phi(x) = \arctan x$. Since ϕ is injective,

$$d(x, y) = 0 \iff \phi(x) = \phi(y) \iff x = y.$$

Symmetry and triangle inequality follow from the usual absolute value metric:

$$|\phi(x) - \phi(z)| \leq |\phi(x) - \phi(y)| + |\phi(y) - \phi(z)|.$$

Exam exercise 2: completeness

Exercise: Question

Is

$$X = \{f \in C[0, 1] : f(0) = 0\}$$

complete with respect to $\|\cdot\|_\infty$?

Hint

View X as a subset of the Banach space $C[0, 1]$. Is it closed?

Solution: completeness of a function subspace

Worked example: Closed subspace argument

Let $f_n \in X$ and $f_n \rightarrow f$ uniformly. Since evaluation at 0 is continuous,

$$f_n(0) \rightarrow f(0).$$

But $f_n(0) = 0$ for all n , so $f(0) = 0$. Hence $f \in X$. Therefore X is closed in $C[0, 1]$, and so it is complete.

Exam exercise 3: projection

Exercise: Question

In $\mathbb{R}^3$, find the projection of

$$x = (1, 2, 3)$$

onto the line spanned by

$$u = (1, 1, 1).$$

Then compute the distance from x to the line.

Solution: projection exercise

$$\text{proj}_u x = \frac{\langle x, u \rangle}{\langle u, u \rangle} u = \frac{1 + 2 + 3}{3} (1, 1, 1) = 2(1, 1, 1).$$

Thus

$$\text{proj}_u x = (2, 2, 2).$$

The residual is

$$r = x - \text{proj}_u x = (-1, 0, 1).$$

The distance is

$$\|r\|_2 = \sqrt{2}.$$

Final checklist for Module 1

Before the exam, be able to do these quickly

- ① Verify metric, norm and inner product axioms.
- ② Decide convergence and Cauchy property in simple examples.
- ③ Classify completeness and compactness in subsets of $\mathbb{R}^n$.
- ④ Use Holder, Minkowski and Cauchy-Schwarz correctly.
- ⑤ Compute kernels, ranges and simple operator norms.
- ⑥ Compute orthogonal projections and interpret residuals.
- ⑦ Produce a counterexample when a hypothesis is missing.

Counterexample bank: metric spaces

Counterexample: Know these by heart

- ① Bounded but not compact: $(0, 1)$ in $\mathbb{R}$.
- ② Closed but not compact: $[0, +\infty)$ in $\mathbb{R}$.
- ③ Cauchy but no limit in the space: rational approximations of $\sqrt{2}$ in $\mathbb{Q}$.
- ④ Continuous but not uniformly continuous: $f(x) = x^2$ on $\mathbb{R}$.
- ⑤ Convergent under one metric but not another: $1/n$ under usual versus discrete metric.

Counterexample bank: normed spaces

Counterexample: Know these by heart

- ① Seminorm not norm: $p(x_1, x_2) = |x_1|$.
- ② Infinite-dimensional norms not equivalent: spikes in $C([0, 1])$.
- ③ Polynomial space not complete under sup norm.
- ④ Derivative unbounded with sup norm on polynomials.
- ⑤ ℓ^1 norm not induced by an inner product: parallelogram fails.

Exam problem 1: metric verification

Exercise: Question

Let $X = \mathbb{R}$ and define

$$d(x, y) = \min\{1, |x - y|\}.$$

Prove that d is a metric.

Solution strategy

Identity and symmetry are immediate. For the triangle inequality, use

$$\min\{1, a + b\} \leq \min\{1, a\} + \min\{1, b\}$$

for $a, b \geq 0$.

Exam problem 2: completeness

Exercise: Question

Let

$$X = \{f \in C([0, 1]) : f(0) = 0\}$$

with the sup norm. Prove that X is complete.

Solution plan

$C([0, 1])$ is complete. Show that X is closed: if $f_n \in X$ and $f_n \rightarrow f$ uniformly, then

$$f(0) = \lim_{n \rightarrow \infty} f_n(0) = 0.$$

A closed subset of a complete space is complete.

Exam problem 3: compactness

Exercise: Question

Let

$$K = \{x \in \mathbb{R}^2 : x_1^2 + x_2^2 \leq 1, x_1 + x_2 \geq 0\}.$$

Is K compact?

Solution

K is the intersection of two closed sets:

$$\{x : x_1^2 + x_2^2 \leq 1\}, \quad \{x : x_1 + x_2 \geq 0\}.$$

It is also bounded by the disk. Therefore it is closed and bounded in $\mathbb{R}^2$, hence compact by Heine-Borel.

Exam problem 4: operator norm

Exercise: Question

Let $T : C([0, 1]) \rightarrow \mathbb{R}$ be

$$T(f) = \int_0^1 tf(t)dt.$$

With the sup norm, show that T is bounded and compute $\|T\|$.

Solution

$$|T(f)| \leq \int_0^1 t|f(t)|dt \leq \|f\|_{\infty} \int_0^1 tdt = \frac{1}{2} \|f\|_{\infty}.$$

So $\|T\| \leq 1/2$. Taking $f \equiv 1$ gives $T(f) = 1/2$ and $\|f\| = 1$, hence $\|T\| = 1/2$.

Exam problem 5: Gram-Schmidt

Exercise: Question

Apply Gram-Schmidt to

$$v_1 = (1, 1, 1), \quad v_2 = (1, 1, 0), \quad v_3 = (1, 0, 0)$$

in $\mathbb{R}^3$.

What will be graded

- Correct first normalization.
- Correct removal of projections.
- Verification of orthogonality.
- Clear final orthonormal family.

Exam problem 6: projection and normal equations

Exercise: Question

Let $H = \mathbb{R}^3$ and approximate $x = (2, 0, 1)$ by vectors in

$$M = \text{span}\{(1, 1, 1), (1, 0, -1)\}.$$

Find the best approximation and the residual.

Method

Let $y = au + bv$. Solve

$$\langle x - y, u \rangle = 0, \quad \langle x - y, v \rangle = 0.$$

Then $y = P_M x$ and $x - y \in M^\perp$.

Exam problem 7: Bessel inequality

Exercise: Question

Let (e_n) be an orthonormal sequence in a Hilbert space. Prove that for every x ,

$$\sum_{n=1}^N |\langle x, e_n \rangle|^2 \leq \|x\|^2$$

for all N , and conclude Bessel's inequality.

Expected proof

Use

$$s_N = \sum_{n=1}^N \langle x, e_n \rangle e_n,$$

show $x - s_N \perp s_N$, then apply Pythagoras.

Exam problem 8: uniform convergence

Exercise: Question

Let

$$f_n(x) = \frac{x}{1 + nx}$$

on $[0, 1]$. Determine whether $f_n \rightarrow 0$ pointwise and uniformly.

Solution

Pointwise: for $x = 0$, $f_n(0) = 0$; for $x > 0$, $x/(1 + nx) \rightarrow 0$. Uniformly:

$$f'_n(x) = \frac{1}{(1 + nx)^2} > 0,$$

so the maximum on $[0, 1]$ is at $x = 1$:

$$\|f_n\|_{\infty} = \frac{1}{1 + n} \rightarrow 0.$$

Thus convergence is uniform.

Exam problem 9: pointwise but not uniform

Exercise: Question

Let

$$g_n(x) = \frac{nx}{1 + n^2 x^2}$$

on $[0, 1]$. Show that $g_n(x) \rightarrow 0$ pointwise, but not uniformly.

Solution

For fixed $x > 0$, $g_n(x) \sim 1/(nx) \rightarrow 0$, and $g_n(0) = 0$. But at $x = 1/n$,

$$g_n(1/n) = \frac{1}{2}.$$

Therefore $\|g_n\|_{\infty} \geq 1/2$, so convergence is not uniform.

Exam problem 10: closed subspace

Exercise: Question

In ℓ^2 , let

$$M = \{x = (x_n) : x_1 = 0\}.$$

Show that M is a closed subspace and compute the projection of x onto M .

Solution

M is linear. If $x^{(k)} \rightarrow x$ in ℓ^2 and each $x_1^{(k)} = 0$, then

$$|x_1| \leq \|x^{(k)} - x\|_2 \rightarrow 0,$$

so $x_1 = 0$. Thus M is closed. Projection:

$$P_M x = (0, x_2, x_3, \dots).$$

Common mistakes in Module 1

Counterexample: Avoid these

- ① Saying “bounded implies compact” outside the exact Heine-Borel setting.
- ② Forgetting that Cauchy sequences may fail to converge in incomplete spaces.
- ③ Treating pointwise convergence as uniform convergence.
- ④ Assuming every norm comes from an inner product.
- ⑤ Applying the projection theorem to a non-closed set.
- ⑥ Calling a linear map continuous without proving a norm bound.

How to write a proof clearly

Four rules

- ① Start by naming the definition or theorem used.
- ② Check hypotheses explicitly.
- ③ Keep estimates visible and justified.
- ④ End with the exact statement required.

Example: A good final sentence

“Therefore every Cauchy sequence in X converges to an element of X , hence X is complete.”

Module 1 summary

Conceptual chain

metric $\Rightarrow$ convergence $\Rightarrow$ Cauchy $\Rightarrow$ completeness

compactness $\Rightarrow$ finite subcovers and extrema

norm $\Rightarrow$ metric plus linear structure

inner product $\Rightarrow$ geometry, angles and projections

Hilbert space $\Rightarrow$ complete geometry

Checklist before moving to Module 2

Exercise: Self-assessment

You should be able to:

- ① Give examples and counterexamples of metric, normed, Banach and Hilbert spaces.
- ② Prove that standard formulas define metrics and norms.
- ③ Use compactness and completeness without confusing them.
- ④ Compute projections and apply the projection theorem.
- ⑤ State and prove Cauchy-Schwarz, Bessel and basic projection identities.
- ⑥ Solve exam-style exercises with clean notation and complete arguments.